SYNTHSTROM DELUGE · RUDEMENT FIRMWARE

Rudement Features

Edition of August 2026

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SYNTHSTROM DELUGE · RUDEMENT FIRMWARE

The

Rudement Features

Operating manuals for the additions in this fork

This compiles a manual for every feature in the Rudement fork that is not part of official or upstream community firmware: the four-band EQ, Bouls, Wisps, the Gristleator, Sear, Kit Split, Aux Sends, and Chroma — plus a MIDI Implementation Chart consolidating every default CC assignment and MIDI-learn behaviour across all of them in one place. Each chapter documents behaviour as built — menu paths, on-screen readouts, defaults, and the handful of known limitations worth reading before judging a feature by an unlucky first setting. Every one of them can be switched off; the first chapter covers how, and what switching one off does and does not change.

EQ4 · BOUL · DUST · GRIS · SEAR · SPLT · BRSH · PREV · RCAP

Edition of August 2026

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SYNTHSTROM DELUGE · RUDEMENT FIRMWARE

The

Community Features

Operating manual for the nine toggles

Every addition in this firmware can be switched off. They live together under `SETTINGS → COMMUNITY FEATURES`, alongside the toggles the official community firmware already ships, and all but one are on when you first boot — the toggle is there to take a feature away, not to opt in to it.

EQ4 · BOUL · DUST · GRIS · SEAR · SPLT · BRSH · PREV · RCAP

Edition of August 2026

WHAT IS ON WHICH BRANCH

No single branch carries all eight. Read down your build's column before you read the chapter.

	clouds-clean-entry 1.2.1	kit-split muted-12 1.2.1	kit-split-13 1.3	aux-sends oled-kitsplit-12 1.2.1	chroma-12 1.2.1
Four-Band EQ	●	●	●	●	○
Bouls	●	●	●	●	○
Wisps	●	●	●	○	○
Gristleator	●	●	●	●	●
Sear	●	●	●	●	○
Kit Split	●	●	●	●	○
Aux Sends	○	○	○	●	○
Chroma	●	●	○	○	●

Aux Sends lives only on the aux-sends branches; Chroma is not on the 1.3 kit-split line. Read from the working tree, not from memory.

Figure 1.1 — What is actually on each branch, read from the working tree. No single branch carries all eight, so find your build's column before you read a chapter.

01 The toggles

WHERE THE TOGGLES LIVE

Every addition in this firmware can be switched off from one place.

SETTINGS › **COMMUNITY FEATURES**

- Four-Band EQ — EQ4
- Bouls Engine — BOUL
- Wisps FX — DUST
- Gristleator — GRIS
- Sear — SEAR
- Kit Split — SPLT
- Chord Brush — BRSH
- Keyboard Note Preview — PREV
- Retrospective Capture — RCAP

Kit Split is the one that ships off, because it is destructive. The other eight are on until you say otherwise.

Figure 1.2 — Where the toggles live, and what each one is called.

FEATURE	7SEG	DEFAULT	SWITCHING IT OFF HIDES
Four-Band EQ	`EQ4`	On	The two mid bands. Bass and treble are stock Deluge and stay exactly where they are, dB and Hz readouts included.
Bouls Engine	`BOUL`	On	`BOULS` in the oscillator type list.
Wisps FX	`DUST`	On	The Wisps menu.
Gristleator	`GRIS`	On	The Gristleator menu.
Sear	`SEAR`	On	The Sear drive and tone controls.

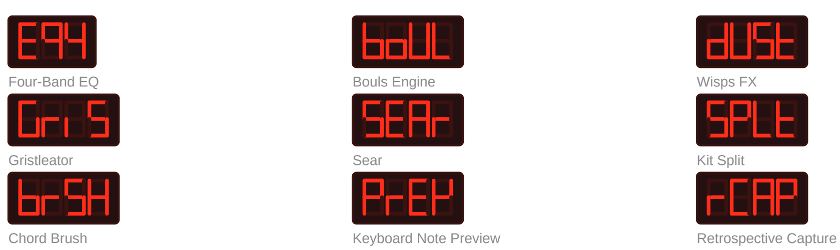
Kit Split	`SPLT`	**Off**	`Kit Global FX → Actions → Split Kit`.
Chord Brush	`BRSH`	On	Chord capture and placement — the Harmonic Brush. The Harmonic layout itself stays where it is.
Keyboard Note Preview	`PREV`	On	The dim highlighting that lights the keyboard grid for sequenced notes during playback. Chapter 09 covers what that buys you on the Harmonic layout.
Retrospective Capture	`RCAP`	On	The SHIFT + RECORD dump of what you just played in Keyboard View.

Kit Split is the one that ships off, because it is destructive: it consumes the kit it is given, and every clip on it. Everything else is additive, so it is on until you say otherwise.

The list sits among the toggles the official community firmware already ships, so these appear alongside entries you may recognise. On a 7-segment display each shows the four-character label in the table. Every one is stored in the persistent settings rather than in the song, so a choice made here follows the instrument rather than the music.

THE NINE TOGGLES ON A 7-SEGMENT DISPLAY

What each one calls itself when there is no OLED to spell it out.



Kit Split is the only one that ships off. Every code here is the string seven_segment.json actually holds.

Figure 1.3 — The same nine on a 7-segment display, drawn with the firmware's own font table — so these are the exact segments the hardware lights.

02 What switching one off does — and does not — do

A toggle hides menus. It does not remove processing.

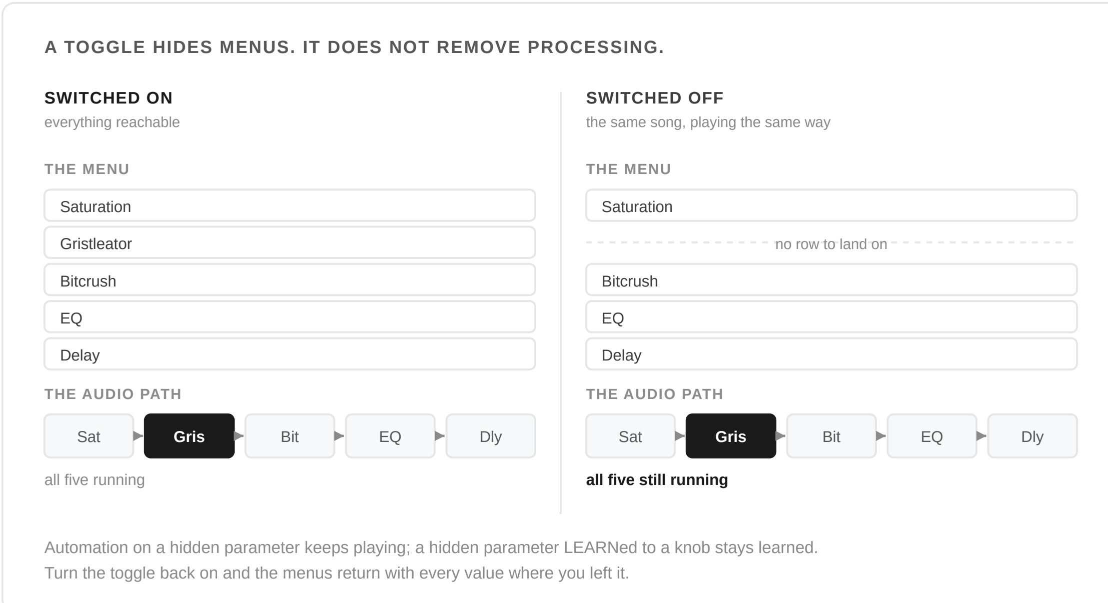


Figure 1.4 — The same song, before and after the Gristleator toggle goes off. One menu row stops being reachable; nothing in the audio path changes. This is the behaviour the stock Enable DX7 Engine toggle already has, followed rather than reinvented.

This matters more than it sounds. If you switch off a feature that a saved song is already using, the song still plays back exactly as saved. A song leaning on the Gristleator keeps chopping; a track sending audio to the CV sockets keeps sending it; a sound built on Bouls keeps sounding like Bouls. What you lose is the ability to reach the controls, not the sound they were set to.

The same holds for everything else those parameters are wired into. Automation lanes on a hidden parameter keep playing. A hidden parameter that has been LEARNed to a knob stays learned. Turn the toggle back on and the menus return with every value where you left it.

> **WHY THEY BEHAVE THIS WAY**
>
> Because the alternative is worse. A toggle that silenced processing would change how an existing song sounds the moment it was flipped — and it would do so on somebody else's machine, when a song written here is opened on a Deluge with different toggles set. Hiding menus while leaving audio alone means a song is a song, wherever it is opened.
>
> This is the behaviour the stock **Enable DX7 Engine** toggle already has. These follow it rather than inventing a second convention.

03 Three that hide a little differently

Four-Band EQ is a partial hide. Only the two mid bells belong to this firmware, so switching it off hides those and leaves the stock bass and treble shelves untouched — along with the change that made all four report in real dB and Hz rather than a bare `0–50`. You are hiding two bands, not reverting the EQ.

Bouls Engine hides a list entry rather than a menu: `BOULS` is simply not offered as an oscillator type. A sound already set to Bouls keeps playing as Bouls — the type just has no row to land on until the feature is switched back on.

Chord Brush hides a gesture rather than a menu. There is nothing to look at either way: switching it off means the SELECT encoder stops arming a chord in a keyboard view and stops stamping one in the piano roll, and both go back to their ordinary jobs. The Harmonic layout, which has no toggle of its own, is untouched.

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SYNTHSTROM DELUGE · RUDEMENT FIRMWARE

The

Four-Band EQ

Operating manual for the two added mid bells

The stock Deluge EQ has two bands: a bass shelf and a treble shelf. This fork adds two bell bands between them — Low mid and High mid — and, at the same time, changes what every EQ control reports. All four amount knobs now read out in dB and all four frequency knobs in Hz, rather than a bare `0–50`.

Bass · Low mid · High mid · Treble

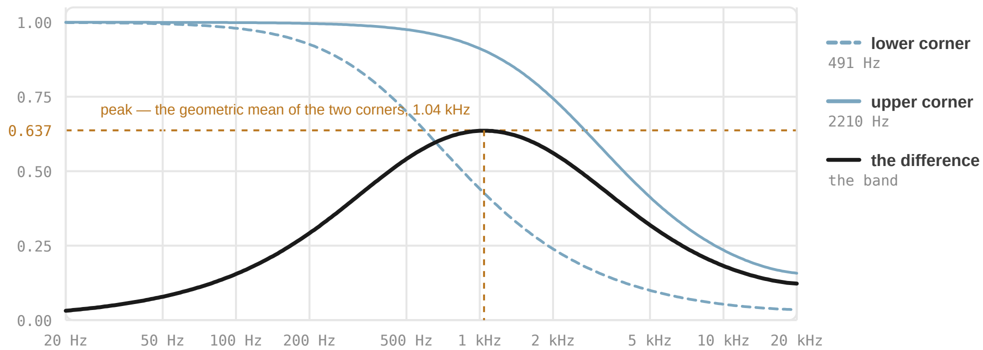
Edition of August 2026

01 What it is

Two bell bands, added between the existing bass and treble shelves. They are not built the way a parametric EQ's bell usually is — there is no biquad peaking filter here. Each band is the *difference* of two one-pole lowpasses, a low corner subtracted from a high corner, which peaks at the geometric mean of the two and falls away on both sides. It behaves like a bell on the knob; structurally it is a subtraction.

A BELL HERE IS A SUBTRACTION, NOT A PEAKING FILTER

Low mid, both Freq knobs at their detent: one lowpass minus a lower one.



It peaks at 0.637 rather than 1.0, so the band never passes the whole of what it extracts — which is why the two mid bands are mixed in at ×6 where the shelves use ×4.

Figure 2.1 — Both curves and the difference between them are computed from the shipped coefficients in `eq_bands.hpp`. It behaves like a bell on the knob; structurally it is a subtraction.

That construction has one consequence worth knowing before you judge the range: a band built this way never passes the full extracted signal at its peak. The low mid captures about 64% of what it pulls out at its neutral centre, the high mid about 60%. Both mid bands are therefore mixed in at ×6 rather than the ×4 the shelves use, which is what gets them to the same useful ±12 dB or so range from a smaller starting capture.

02 Finding it

FINDING THE EQ

Four bands, eight controls. Menu position 25 is the detent on every one.

FX › **EQ**

Bass amount	BASS	High mid amount	HIGH MID
Bass freq	Bfrq	High mid freq	HMfq
Low mid amount	LOW MID	Treble amount	TREBLE
Low mid freq	LMfq	Treble freq	Tfrq

The two mid bells are this firmware's. Bass and treble are stock Deluge and stay where they are when the toggle goes off.

Figure 2.2 — The EQ's eight controls, and where to find them.

Your first minute

#	DO THIS	AND WATCH
1	Put a bright loop on a track and open `FX → EQ`	Eight controls, four bands. Every one reads in real dB or Hz, not 0–50
2	Take **Low mid amount** to the bottom	The popup reads `OFF` rather than a number — the band has cancelled, and a floor is more honest than a rising negative
3	Bring it back to 25	`0.0 dB`. Centre is the detent on all eight controls
4	Sweep **Low mid freq** up past knob 41	The readout stops moving. The upper pole has pinned — that is the band"s range, not a stuck encoder
5	Now cut instead of boosting	A cut goes markedly deeper than a boost reaches. That asymmetry is the amount law, and it is the fastest way to hear where a band actually sits

The EQ occupies the same slot stock Deluge EQ always has, late in the global FX chain.

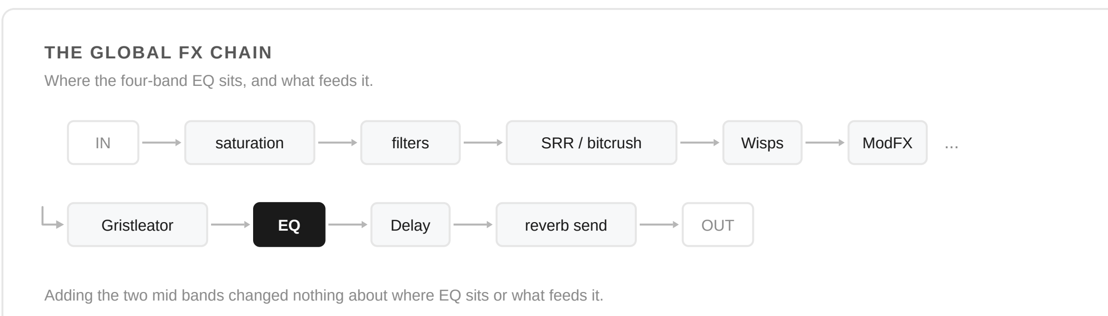


Figure 2.3 — The EQ occupies the same slot stock Deluge EQ always has, late in the global FX chain. The wrap is a line break, not a feedback path.

Adding the two mid bands changed nothing about where EQ sits or what feeds it — they were inserted into the same processing block the bass and treble shelves already ran in.

03 The controls

Menu position 25 is centre / the detent on every one of these eight controls. Figures below were checked against the same fixed-point law the audio path runs, not just the nominal design values.

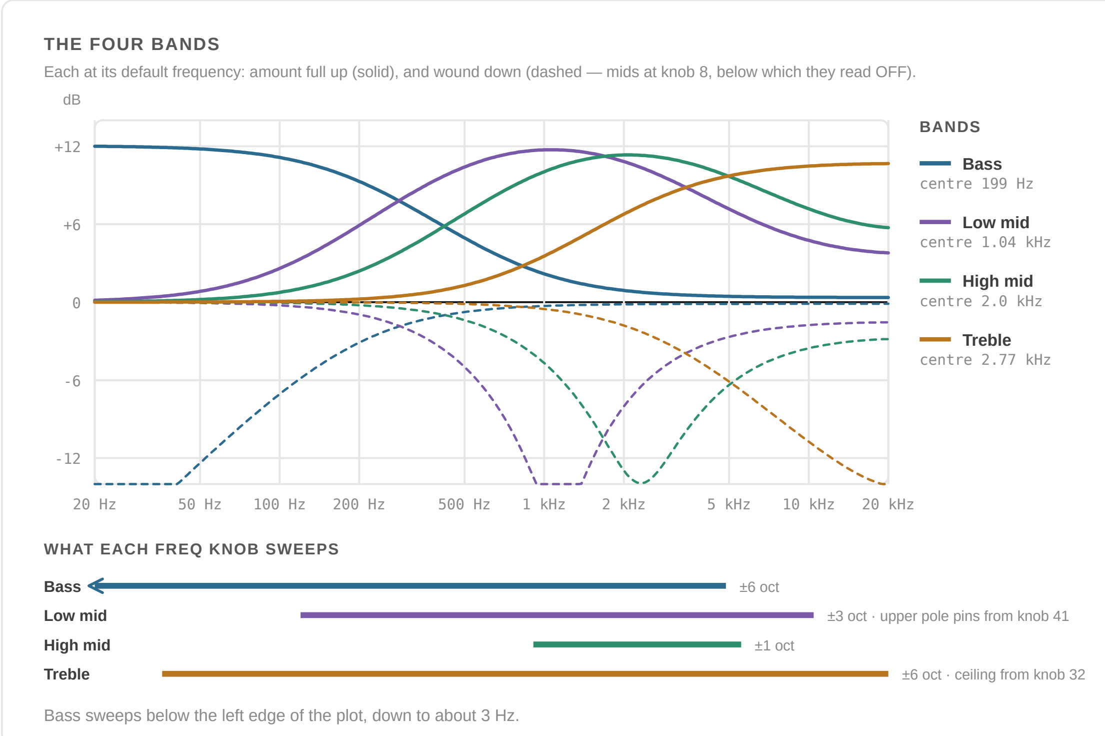


Figure 2.4 — Plotted from the same fixed-point law the audio path runs, so every centre frequency and every gain here is the number the menu reports. Note how much deeper a cut goes than a boost: that asymmetry is the amount law, not the plot.

Bass & Treble (the shelves)

KNOB	BASS CORNER	TREBLE CORNER
0	~3 Hz	~36 Hz
25 (centre)	~199 Hz	~2.77 kHz
32	~658 Hz	20 kHz (ceiling)
50	20 kHz (ceiling)	20 kHz (ceiling)

Both shelves sweep ±6 octaves around their centre corner. Amount is an exact ±12 dB at the top/bottom of its own knob, true zero (a flat line) at the bottom of Bass. Treble's Freq knob reaches the display ceiling of 20 kHz by knob position 32 — and from there up, the Treble *amount* knob does nothing at all, because the shelf has nothing left above Nyquist to remove. That is not a bug in the readout; it is the digital filter reporting the truth about itself.

Low Mid & High Mid (the bells)

KNOB (FREQ)	LOW MID CENTRE	HIGH MID CENTRE
0	~120 Hz	~910 Hz
25 (centre)	~1.04 kHz	~2.0 kHz
41	~6.4 kHz (upper pole pinning)	—
50	pinned near 10–20 kHz	~5.6 kHz

KNOB (AMOUNT)	LOW MID GAIN	HIGH MID GAIN
0	OFF*	−20.0 dB
12	−11.5 dB	−10.2 dB
25 (centre)	0.0 dB	0.0 dB
38	+7.0 dB	+6.8 dB
50	+11.7 dB	+11.4 dB

* Below −24 dB the popup reads `OFF` rather than a number — the bells actually cancel a hair past zero at the bottom of their travel, and a display floor is more honest than a negative-then-rising number as you wind the knob back up.

KNOWN LIMITATIONS

Low mid's upper pole pins from around Freq knob 41 — above that, moving the knob further doesn't move the band, it's already sitting at the display ceiling.

Treble's amount does nothing above Freq knob 32, for the same reason: nothing left to shelve above Nyquist.

Do not raise either mid band's octave sweep beyond its shipped range (Low mid 3 octaves, High mid 1). At higher values both of a bell's one-poles saturate at once, their difference cancels, and the band goes silent over the top of its own knob — this is stated directly in the source and is not a hypothetical.

04 Reference

THE EQ, FOUR CHARACTERS AT A TIME

Amount and frequency for each of the four bands.

Figure 2.5 — All eight, as a 7-segment display renders them.

CONTROL	7SEG	DEFAULT
Bass amount / freq	`BASS` / `Bfrq`	centre (0 dB) / ~199 Hz
Treble amount / freq	`TREBLE` / `Tfrq`	centre (0 dB) / ~2.77 kHz
Low mid amount / freq	`LOW MID` / `LMfq`	centre (0 dB) / ~1.04 kHz
High mid amount / freq	`HIGH MID` / `HMfq`	centre (0 dB) / ~2.0 kHz

> **SWITCHING IT OFF ENTIRELY**
>
> `SETTINGS → COMMUNITY FEATURES → Four-Band EQ` (`EQ4`), on by default. Switching it off hides **only** the two mid bands. Bass and treble are stock Deluge and stay exactly where they are, dB/Hz readouts included — this toggle does not touch them.

CONTROL	HOW THE READOUT IS DERIVED
Bass	Exact, and needs nothing but the amount — a lowpass is exactly 1 at DC, so the shelf's own gain law is the whole answer.
Treble	Reads its own Freq param as well as the amount, because a digital lowpass is nowhere near 0 at Nyquist — ignoring that would report gain the band cannot actually reach.
Low mid / High mid	Take a fixed capture at the band's neutral centre rather than tracking Freq, to stay monotonic through the region where the amount law over-cuts into polarity inversion.

PROVENANCE

Bass and treble shelves are stock Deluge. The two mid bells, and the Hz/dB readout on all four bands, are original to this fork's 1.2.1 line and were carried unchanged to 1.3. Source: `src/deluge/gui/menu_item/eq/`, `src/deluge/dsp/eq_bands.hpp`.

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SYNTHSTROM DELUGE · RUDEMENT FIRMWARE

The

Bouls

Operating manual for the 24-model oscillator engine

Mutable Instruments Plaits, as a Deluge oscillator type. Twenty-four synthesis models behind one front panel — Harmonics, Timbre and Morph — plus a low-pass gate and a choice of which of the engine's two outputs to take. The three controls keep their names in every model and change their meaning in every model; sections 03 to 07 are about what they actually do, model by model.

Model · Harmonics · Timbre · Morph · LPG

Edition of August 2026

01 What it is

Bouls is available as an oscillator type: select it, and the sound uses whichever of its 24 synthesis models is dialled in instead of a conventional waveform. It is available on **oscillator 1 only, and not in kits** — the same restriction the DX7 engine has.

Harmonics, Timbre and Morph are not new parameters bolted on beside the existing oscillator controls — they reuse three slots that already existed for other oscillator types (pulse-width, wave index, and a second pulse-width), running at the model's own front-panel meaning instead. That means all three patch, modulate and automate exactly the way pulse-width already did for a square wave, with no new patching surface to learn.

Three things decide what you hear: **which model** is selected, **where the three controls sit** within that model, and **whether the low-pass gate is in circuit**. The third is the one people miss. With the LPG off, Bouls is a bare oscillator and the Deluge's own envelopes and filter do all the shaping; with it on, the engine gets the struck, blooming, vactrol-flavoured envelope that most of the module's recognisable sounds are made of. Section 04 covers what turning it on changes — which is more than an envelope, on four of the models.

02 Finding it

FINDING BOULS

Select it as the oscillator type, then press SELECT on it again.

`Osc 1` › `type` › **`BOULS`** oscillator 1 only, and not in kits

Bouls model	`MODL`	Low-pass gate	`LPG`
Harmonics	`HARM`	LPG decay	`DECY`
Timbre	`TIMB`	LPG colour	`COLR`
Morph	`MORF`	Aux output	`AUX`

None of these cut a held note when changed — the voice re-reads model, LPG and aux output every render block.

Figure 3.1 — The Bouls menu, and the two presses that reach it.

Your first minute

#	DO THIS	AND WATCH
1	Set oscillator 1"s type to `BOULS`, then press SELECT on it again	The Bouls menu: model, three controls, the gate, and the aux output
2	Put **Harmonics**, **Timbre** and **Morph** at about 25 each	A fresh sound sits at 0 on all three, which is where several models are dull, thin or nearly silent. This is a better place to listen from
3	Leave the model at 8 and sweep **Timbre**	Virtual Analog, the plainest voice in the list. The square fades in, narrows, then hard-syncs past the middle
4	Turn **Low-pass gate** on	The same oscillator becomes a struck, blooming thing. This is the setting most of the recognisable sounds are made of
5	Change the model to 21 — Bass Drum — and take **Morph** up	Nothing until you do: Morph is the decay on the drums, and it defaults to 0, which is why an untouched Bass Drum is all click

Select `BOULS` as the oscillator type, then press select on it again to enter the Bouls menu: **Bouls model**, **Harmonics**, **Timbre**, **Morph**, **Low-pass gate**, **LPG decay**, **LPG colour**, and **Aux output** (which of the engine's two outputs you take). None of the toggles cut a held note when changed — the voice re-reads model, LPG and aux output every render block, so switching any of them mid-note swaps live rather than retriggering. The module itself isn't click-free when you do this either, so a small seam on model changes is expected, not a fault.

The three continuous controls in this menu are the same parameters as Osc 1's Pulse Width and Wave Index and Osc 2's Pulse Width — a second door onto one room, not a copy. Editing them here, on the oscillator menus, from the shortcut grid, over MIDI or from automation are all the same edit.

03 Harmonics, Timbre and Morph

Three controls, twenty-four meanings each. The menu never renames them per model — it says Harmonics, Timbre and Morph whichever model is loaded — so section 07 is the translation table, and it is the part of this chapter worth keeping open.

Loosely, and with plenty of exceptions: **Harmonics** chooses *what* — the chord, the terrain, the patch, the filter mode, the spectrum's shape. **Timbre** is usually the brightness or density control, the one that behaves most like opening a filter. **Morph** is usually the waveform or the decay — the shape of the thing rather than its content. Learning a model means learning which of those three the model bent furthest.

Travel, and where the useful part of it is

CONTROL	BORROWED SLOT	MENU DEFAULT	WORKING TRAVEL
Harmonics	Osc 1 Pulse Width	0	0 → 50 covers the model's full range
Timbre	Osc 1 Wave Index	0	0 → 50 covers the model's full range
Morph	Osc 2 Pulse Width	0	0 → 50 covers the model's full range

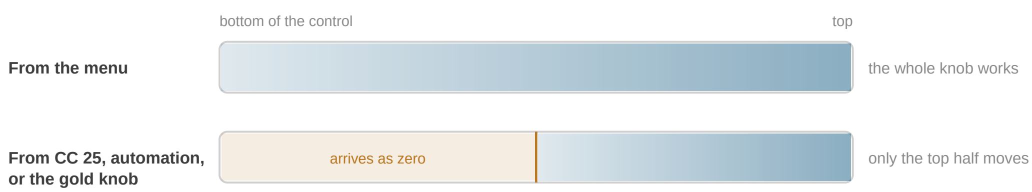
All three menus run 0 to 50 across the engine's full 0–1, and a fresh Bouls sound sits at **0 on all three** — which is the engine's zero on all three, where several models are dull, thin or nearly silent. *If a model sounds like nothing, that is the first thing to check.* Putting all three at about 25 lands you in the middle of every model's range, and is a better place to start listening from than the default.

Timbre needs a word of explanation, because the slot underneath it does not behave like the other two. Harmonics and Morph borrow pulse-width parameters, which run from zero upwards. Timbre borrows the wave-index parameter, which is *centred*: its middle is zero and its lower half is negative, and the engine has no use for negative values, so everything below the middle arrives as zero. Until August 2026 that was exactly what you got — a Timbre control that did nothing below 25 and had half the resolution of the other two above it. The two menus that reach this parameter (Timbre in the Bouls menu, and Osc 1's wave position, which renames itself Timbre for a Bouls sound) now scale it like a pulse width, so the whole knob works. Saved sounds are unaffected: the stored value and the engine mapping are both unchanged, so an old sound plays back exactly as it did — only the number the menu shows for it has moved.

Modulation is clamped the same way: a patch cable, an LFO or an envelope that pushes any of the three below the engine's zero is held at zero rather than wrapping. Modulating any of them up from a low setting therefore only ever moves them upward.

TIMBRE, AND THE SLOT UNDERNEATH IT

Harmonics and Morph borrow pulse-width params, which run from zero upwards. Timbre borrows the wave-index param, which is centred: its middle is zero and its lower half is negative, and the engine has no use for negative values.



The two menus that reach this parameter now scale it like a pulse width; saved sounds are unaffected, since only the number the menu shows has moved. Everything else still writes the parameter directly, across its centred range.

Figure 3.2 — Why Timbre behaves differently depending on where you reach it from. The menus were re-scaled; nothing else was.

TIMBRE, FROM ANYWHERE OTHER THAN THE MENU

Only the two menus were re-scaled. **MIDI CC 25, the Automation View lane and the gold knob write to the parameter directly**, across its centred range — so from those three, Timbre's lower half still lands on the engine's zero and the control only starts moving from its midpoint. A CC 25 sweep from 0 to 127 gives you the model's full Timbre range across the top half of the fader; the bottom half is silent travel.

KNOWN LIMITATIONS

Morph borrows **Oscillator 2's** pulse width. If oscillator 2 is set to a square or pulse wave, its width and Bouls' Morph are the same control and cannot be set independently. Harmonics and Timbre are safe — they sit on oscillator 1's slots, which Bouls has taken over anyway.

Model, Low-pass gate, LPG decay, LPG colour and Aux output are per-source settings rather than parameters: they are saved with the sound, but they cannot be automated, patched or MIDI-learned.

04 The low-pass gate, and what turning it on changes

On the module the LPG is in circuit unless you patch the level input, and it supplies most of what people recognise as "Plaits" — the pluck, the bloom, the way a bright attack darkens as it falls away. Here it is a toggle, off by default, because a Deluge oscillator is expected to be a bare oscillator until told otherwise.

Low-pass gate default off

Off: the engine's raw output, no gate and no decay. The Deluge's own envelopes, filter and amplitude do all the work, and Bouls behaves like any other oscillator type.

On: the gate is struck on each note-on and decays under LPG decay and LPG colour. Its attack scales with pitch, so low notes bloom noticeably more slowly than high ones. This is the setting for plucks, bongos, blooming pads and anything that should have a shape of its own before the Deluge's envelope touches it — and it stacks with that envelope rather than replacing it, so a slow Deluge attack over a struck LPG gives you both.

LPG decay default 25

The gate's decay time, from a click of a few milliseconds at 0 to a tail of well over a second at 50. It has no effect while the low-pass gate is off, and none on the eight self-enveloped models below.

LPG colour default 25

What kind of gate it is. At 0 it is a filter: the sound darkens as it decays, the vactrol behaviour. At 50 it is a plain amplitude gate — the tone holds and only the level falls. It also lengthens the tail slightly as it rises. Only read while the low-pass gate is on.

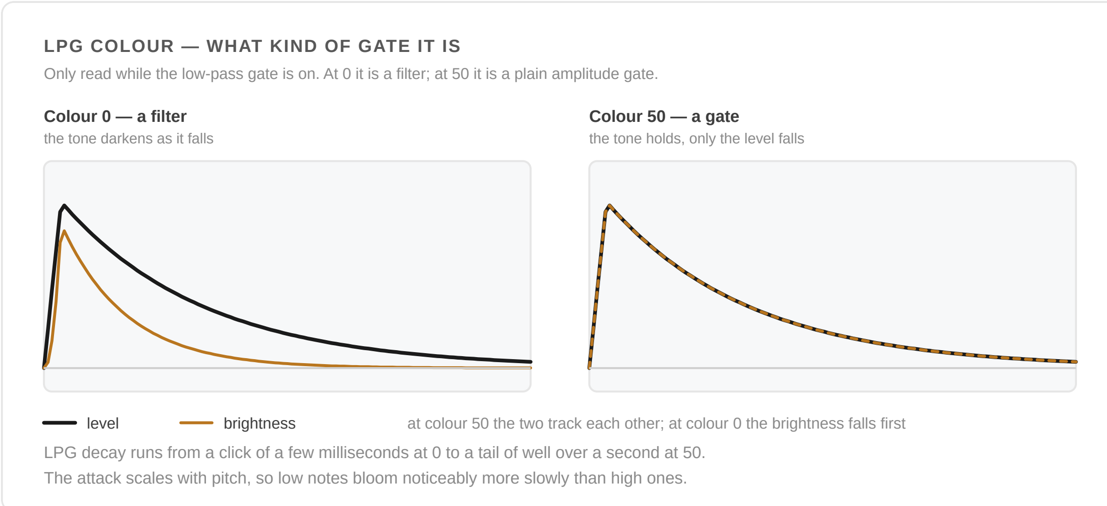


Figure 3.3 — LPG colour, at both ends of its travel. At 0 the brightness falls faster than the level — the vactrol behaviour; at 50 the two fall together and the tone holds.

The eight models that bring their own envelope

Eight models declare their own amplitude shaping, and the engine bypasses the LPG for them regardless of the toggle: the three **6-op FM** banks, **Inharmonic String**, **Modal Resonator**, **Bass Drum**, **Snare Drum** and **Hi-Hat**. On those eight, the Low-pass gate toggle, LPG decay and LPG colour are all inert — their decay lives elsewhere: on **Morph** for the three drums, the string and the resonator, and inside the loaded patch for the FM banks. This is upstream's design, not a limitation of the port, and it is not optional: without it the drums would have nothing to strike them and would sit silent.

The side effect: four models change behaviour, not just shape

Turning the LPG on also hands the engine a trigger, and a handful of models read a trigger as a mode switch rather than as an envelope. On these, the LPG toggle is doing something bigger than adding a decay:

- **Chiptune** — stops playing a chord and becomes an *arpeggiator*, advancing one step per note-on, with Timbre selecting the pattern instead of the inversion. The gate itself is bypassed, so you get the mode change without the decay.
- **Granular Cloud** — the swarm stops free-running and fires a fresh burst of grains at each note-on.
- **Filtered Noise** — the noise clock's range extends two octaves lower, and the clock resets at each note-on rather than drifting.
- **Speech** — in the word banks, each note-on replays a whole word instead of the word being scrubbed by hand with Morph.

Particle Noise and **Filtered Noise** also reseed their randomness on each note-on once the trigger is live, which makes repeated notes noticeably more consistent than they are with the LPG off.

05 The aux output

ONE ENGINE, TWO OUTPUTS — YOU TAKE ONE OF THEM

Both are computed either way, so the toggle costs nothing and can be flipped on a held note.

what the aux gives you, by kind of model

· a second oscillator · the exciter before a resonator · a sub · a bit-crushed copy · the other side of a stereo pair

On the three 6-op FM banks the aux is identical to the main, so the toggle does nothing there.

Levels differ markedly between the two on some models — expect to re-balance the track's volume after switching.

Figure 3.4 — Every model renders two signals at once. Aux output decides which of them the voice takes — it is a choice, not a mix.

Every model renders two signals at once. The main output is the model's headline sound; the aux output is a companion the same engine produced along the way — a second oscillator, the exciter before a resonator, a sub, a bit-crushed copy, or the other side of a stereo pair. **Aux output** chooses which of the two the voice takes.

Both are computed either way, so the toggle costs nothing and can be flipped on a held note. It is not a mix and there is no blend control: you get one or the other. To hear both at once you need two tracks, since Bouls is oscillator 1 only.

What aux gives you is listed per model in section 07, and gathered in the table in section 08. Two notes worth having up front: on the **three 6-op FM banks the aux output is identical to the main one**, so the toggle does nothing there; and levels differ markedly between the two outputs on some models, so expect to re-balance the track's volume after switching.

06 The 24 models

THE TWENTY-FOUR MODELS, AND WHAT THE LPG DOES TO EACH

Indices 0–7 are the eight models Plaits 1.2 added; 8–23 are the original sixteen, which is why the classic Virtual Analog sits at 8.

#	MODEL	7SEG	LPG
00	Virtual Analog VCF	VAVC	•
01	Phase Distortion	PHDS	•
02	6-op FM A	FMA	▬
03	6-op FM B	FMB	▬
04	6-op FM C	FMC	▬
05	Wave Terrain	TERR	•
06	String Machine	STRM	•
07	Chiptune	CHIP	○
08	**Virtual Analog** default	VANA	•
09	Waveshaping	WSHP	•
10	2-op FM	FM2	•
11	Granular Formant	FORM	•
12	Harmonic	HARM	•
13	Wavetable	WTBL	•
14	Chords	CHRD	•
15	Speech	SPCH	○
16	Granular Cloud	CLUD	○
17	Filtered Noise	NOIS	○
18	Particle Noise	PART	•
19	Inharmonic String	STRG	▬
20	Modal Resonator	MODL	▬
21	Bass Drum	BASS	▬
22	Snare Drum	SNAR	▬
23	Hi-Hat	HHAT	▬

▬ brings its own envelope — the LPG toggle, decay and colour are all inert

○ reads the LPG's trigger as a MODE CHANGE, not just an envelope — see §04

• ordinary: the LPG adds a struck, blooming envelope and nothing else

Figure 3.5 — The whole list, with what the low-pass gate does to each. Eight models bring their own envelope and bypass the LPG whatever the toggle says; four more read its trigger as a mode change rather than as a shape. The list order is fixed — a saved sound stores the raw index.

Indices 0–7 are the eight models Mutable added in Plaits firmware 1.2; 8–23 are the original sixteen, which is why the "classic" Virtual Analog sits at index 8 and is the default. The list order is fixed — a saved sound stores the raw index, so re-sorting the list would silently repatch every song that uses Bouls. The figure above is the list: number, name, four-character code, and what the low-pass gate does to each.

07 The models, one by one

Each entry describes the voice, then what the three controls do to it, then what the aux output gives you instead. Where a control behaves differently either side of its centre, that is said — several of them are bipolar in effect even though the menu reads 0–50.

00 · Virtual Analog VCF `VAVC`

A saw-or-pulse oscillator and a sub-oscillator through a resonant low-pass filter with soft clipping at every stage. The only model with a filter of its own, and the one that can sound like a finished patch before the Deluge's filter touches it — a mono synth in a single oscillator slot.

Harmonics resonance, and the filter's slope. The centre is clean; resonance climbs toward both ends and screams at either extreme. The lower half runs both filter stages for a steeper, more polite four-pole; the upper half fades the second stage out for a rawer two-pole with more drive.

Timbre cutoff, ten octaves of it, tracking the note.

Morph the waveform: saw at the bottom, through square, to a narrow pulse at the top — with a sub-oscillator fading in at both extremes and absent in the middle.

Aux the filter's high-pass output instead of its low-pass: thin, buzzy, and useful under something else.

01 · Phase Distortion `PHDS`

Casio CZ-style phase distortion: a sine wave read through a warped phase, which produces filter-like sweeps and hollow resonant vowels with no filter anywhere in the signal path.

Harmonics the ratio between the carrier and the modulator that warps it, quantised to musical intervals rather than swept continuously.

Timbre how much distortion. At zero this model is a plain sine.

Morph the asymmetry of the warp — the pulse width of the distortion, from symmetrical to sharply skewed.

Aux the same shaping free-running instead of hard-synced: phase *modulation* rather than phase distortion, less vocal and more bell-like.

02–04 · 6-op FM A, B and C `FMA` `FMB` `FMC`

Three banks of thirty-two six-operator FM patches — the DX7 architecture, complete with each patch's own operator envelopes and LFO. The three banks are three different sets of thirty-two, selected as three separate models rather than by a bank control.

Harmonics patch select within the bank: thirty-two of them, stepped with hysteresis so a wobbling modulator does not flicker between neighbours.

Timbre brightness — it scales the operators' output levels, which is FM's equivalent of opening a filter.

Morph envelope control. At 0 the patch's own envelopes run at their snappiest — short and percussive; raising it stretches them out toward their full length.

Aux identical to the main output. The toggle does nothing on these three.

Self-enveloped: the LPG toggle, decay and colour are all inert; the patch's envelopes follow the note instead. Two voices are rendered round-robin inside the engine, so a fast retrigger does not cut the previous note's tail. The banks are the factory ones — the module's ability to load your own SysEx bank from a card is not wired up here.

05 · Wave Terrain `TERR`

An orbit traced across a three-dimensional landscape, where the height under the pointer is the waveform. Tight orbits stay near a sine; wide ones cross ridges and gullies and turn spiky and complex.

Harmonics which terrain — eight of them, crossfaded rather than switched.

Timbre the radius of the orbit: how much of the terrain the trace covers, and so how far from a sine it gets. It is automatically reined in on high notes.

Morph the orbit's offset across the terrain — where the loop sits, and how lopsided it becomes.

Aux a sine reading of the orbit summed with the terrain height: a softer, more harmonic relative of the same motion.

06 · String Machine `STRM`

Four divide-down organ voices playing a chord, through a low-pass filter and an ensemble chorus. A Solina in a box, and the most immediately usable pad in the list.

Harmonics the chord, from the eleven-chord bank in section 08.

Timbre filter cutoff and the ensemble together: brightness rises across the whole travel, while the chorus is at its strongest at both ends and falls to nothing in the middle.

Morph the registration — drawbars, in effect. Square at the bottom, through saw and stacked saws, to a bright full registration with upper harmonics at the top.

Aux the other side of the ensemble's stereo pair: the same chord, different chorus phase.

07 · Chiptune `CHIP`

Four square-wave voices and a bass note, deliberately crude: a 1980s console sound chip, phase-inverted voice by voice so the chord fizzes the way those chips did.

Harmonics the chord.

Timbre the inversion — which note of the chord sits on top. With the LPG on, it selects the arpeggiator pattern instead.

Morph pulse width, from a full square to a thin, nasal buzz.

Aux the bass square alone, an octave below the root.

With the low-pass gate on, this model becomes an arpeggiator: one note per note-on, stepping through the chord, with the gate itself bypassed. That is a mode change rather than an envelope — see section 04.

08 · Virtual Analog `VANA` default model

Two oscillators — a variable-width square and a variable-shape saw — mixed together. The plainest and most predictable voice in the list, and the right one for learning what the three controls feel like.

Harmonics detuning between the two oscillators, from unison to a wide beat.

Timbre the square: its level fades up from silence at the very bottom, then its pulse width narrows, and past the middle it hard-syncs, sweeping the classic sync tear.

Morph the saw: shape and width, with its level falling away as the control rises, so the top of the travel is square-dominant and the bottom is saw-dominant.

Aux "monster sync" — both oscillators self-synced under Timbre, harsher and more metallic than the main mix.

09 · Waveshaping `WSHP`

A band-limited slope through a waveshaper and then a wavefolder — the West Coast timbre, where a control that looks like a level control turns into a brightness control.

Harmonics which waveshaping curve, interpolated across a table. The centre is neutral; both directions bend the curve further, and the range is automatically narrowed on high notes to keep the folding from aliasing.

Timbre wavefolder drive: how many folds, and so how many overtones. At 0 there is no folding at all and the model sounds nearly inert — this is the model most often written off from its default.

Morph the slope's shape and pulse width: the asymmetry going into the shaper.

Aux a sine crossfaded with a second fold curve — the same gesture with the top taken off, and a good deal easier on the ears.

10 · 2-op FM `FM2`

One sine modulating another, plus a sub-oscillator — the two-operator FM from Braids, Rings and Elements. Bells, basses, clangs and electric pianos.

Harmonics the modulator's frequency ratio, quantised to a table of musical ratios, so it lands on intervals rather than sliding between them.

Timbre the FM index — the amount of modulation, and therefore the brightness. It is automatically tamed on high notes to hold aliasing down.

Morph feedback, bipolar: the centre is clean, below it the feedback goes to the carrier's phase, above it to the modulator's. The two directions sound quite different — one grows a rasp, the other a growl.

Aux the sub-oscillator, an octave down and itself modulated by the carrier: a fatter, darker version of the same event.

11 · Granular Formant `FORM`

Two grain oscillators whose windows produce a formant peak — a vowel-like, VOSIM-flavoured sound that is pitched by the note and coloured by where the formant sits.

Harmonics the ratio between the two formants, spread over four octaves; below the centre it also bleeds the raw carrier back in, which thickens the low end.

Timbre the formant frequency itself — the vowel, in effect, sweeping seven octaves.

Morph the shape of the grain window, from soft to sharply pinched.

Aux a different formant oscillator at the same pitch: one cleaner peak instead of two interfering ones.

12 · Harmonic `HARM`

Additive synthesis: twenty-four harmonics whose amplitudes are drawn by three controls rather than twenty-four faders. Organs, bells, and spectra that no filter would produce.

Harmonics the number of peaks in the spectrum, from a single broad bump up to sixteen — one peak is a formant, many is a comb.

Timbre where the peak sits: which harmonic is loudest. Sweeping it is the closest thing here to a band-pass sweep.

Morph the slope — how sharply the amplitudes fall away from the peak. Shallow slopes are bright and buzzy, steep ones nearly sinusoidal.

Aux eight organ harmonics — octaves and fifths only — instead of the full integer series: a drawbar organ standing next to a spectrum.

13 · Wavetable `WTBL`

A grid of waves read as a smooth surface, with two axes of position and a third of bank. Sweeping any of the three is the sound of this model.

Harmonics the bank. Past the middle of its travel the interpolation between waves is progressively quantised, and the model turns deliberately steppy and lo-fi — that crunch is the design, not a fault.

Timbre position along a row: eight waves, crossfaded.

Morph position down the column: eight rows, crossfaded.

Aux the same wave crushed to five bits — a dirtier twin of whatever the main output is doing.

14 · Chords `CHRD`

Four notes from a bank of eleven chords, played by divide-down organ voices that hand over one at a time to a wavetable of string and vocal waves at the top of Morph. One key, one chord, and a bass note on the aux output.

Harmonics the chord, stepped through the eleven-chord bank in section 08.

Timbre the inversion: it rotates the voicing upward one note at a time, crossfading rather than jumping, which makes it usable as a modulation destination.

Morph the tone. The bottom is the fullest, brightest organ registration, thinning through saws to a plain square by about the middle; past that the four notes cross one by one into the wavetable and the chord turns to strings and voices.

Aux the root note alone. This is how one voice gives you a bass line and a chord — on two tracks, or by switching.

15 · Speech `SPCH`

Three speech synthesisers stacked behind one control: a naive formant filter, a re-creation of SAM (the 1982 home-computer voice) and an LPC synthesiser with banks of stored words.

Harmonics which synthesiser, and what it can say. The first sixth of the travel crossfades the naive formant filter into SAM; the next sixth crossfades SAM into LPC vowels; the remaining two-thirds step through five banks of LPC words — numbers, letters and so on.

Timbre formant shift: the size of the speaker, from a giant down to a chipmunk, without changing pitch.

Morph the address — which phoneme, vowel or word. Sweeping it by hand is how you get the talking-through-a-scrubbed-tape effect.

Aux the excitation alone: the glottal buzz before the formants, which is a usable raw source in its own right.

With the low-pass gate on, the word banks replay a whole word at each note-on instead of being scrubbed by Morph.

16 · Granular Cloud `CLUD`

A swarm of detuned voices doing random walks around the note — less a chord than a cloud, and the model that best rewards a slow modulation on all three controls at once.

Harmonics the spread: how far the swarm wanders from the root, from a tight unison shimmer to a smear across octaves.

Timbre the density and rate of the swarm's motion.

Morph grain duration — long grains hold pitch, short ones turn the cloud into texture.

Aux the same swarm rendered as sines rather than saws: softer, more choral, far less harsh at high densities.

With the low-pass gate on it switches to burst mode: each note-on fires a fresh burst of grains rather than running continuously.

17 · Filtered Noise `NOIS`

Clocked noise through a resonant multimode filter. It covers everything from wind and surf to a pitched whistle, and it is the model to reach for when a track needs air rather than notes.

Harmonics two things at once: the filter mode, sweeping continuously from low-pass through band-pass to high-pass, and the pitch offset of the second band-pass that appears on the aux output.

Timbre the noise clock. At the bottom the noise is slow, stepped and gritty; at the top it is full-band hiss.

Morph resonance, from a broad wash up to a self-oscillating whistle at the note's pitch.

Aux two band-pass filters summed instead of the single multimode filter: a two-peaked, more vocal noise.

With the low-pass gate on, the noise clock's range extends two octaves lower and resets at each note-on.

18 · Particle Noise `PART`

Random pulses fired through randomly-tuned band-pass filters: rain, crackle, Geiger counters, and — at high density with resonance up — something close to a struck metal sheet.

Harmonics how widely the filter frequencies scatter around the note, from all-in-tune to four octaves of confetti.

Timbre density: how often a particle fires, from occasional ticks to a continuous rustle.

Morph split at its centre. Below it, a diffuser smears the particles into a cloud, most heavily at the bottom; above it the band-passes get resonant and each particle rings instead of clicking.

Aux the raw pulses before any filtering: bare clicks rather than pings.

19 · Inharmonic String `STRG`

A plucked string model with a stiffness term, so it runs from nylon through piano wire to things that stopped being strings some way back.

Harmonics inharmonicity. There is a deliberate dead zone right at the centre where the string is clean and harmonic; move either way and it grows progressively more dispersed and bell-like, with the two directions taking different routes there.

Timbre the brightness of the pluck — the filter on the excitation burst, from a soft thumb to a hard plectrum.

Morph damping: how long the string rings after the pluck.

Aux the raw excitation burst without the string — a short filtered noise click.

Self-enveloped: the LPG toggle and its decay and colour are inert, and Morph is the decay control. The string is always plucked here — the module's continuously-excited, bowed behaviour needs a level input the Deluge does not provide.

20 · Modal Resonator `MODL`

A bank of tuned modes struck by an impulse: bells, bars, bowls, tubes and gongs. The same family as the string above, with the resonator's tuning opened up instead of its stiffness.

Harmonics the inharmonicity of the mode series — harmonic and string-like at one end, sharply metallic at the other. It is deliberately smoothed, so sweeping it glides rather than steps.

Timbre the brightness of the strike: how much energy goes into the upper modes.

Morph damping — a dead thud at the bottom, a long ring at the top.

Aux the exciter alone, without the resonator.

Self-enveloped, exactly as the string: LPG inert, Morph is the decay, always struck rather than continuously excited.

21 · Bass Drum `BASS`

Two bass drums rendered at once: an 808-style bridged-T circuit on the main output, and a synthetic, FM-flavoured drum on the aux.

Harmonics the punch, in three stages as it climbs: attack FM first, then self-FM, then overdrive.

Timbre tone — the balance of click and body.

Morph decay, from a tight tick to a long boom.

Aux the synthetic drum instead of the analogue one: tighter, more modern, more distorted.

Self-enveloped. LPG decay does nothing here — **Morph** is the decay control, and it defaults to 0, which is why an untouched Bass Drum is all click and no drum.

22 · Snare Drum `SNAR`

An 808-style analogue snare — two tuned resonators and a noise burst — on the main output, with a synthetic snare on the aux.

Harmonics snappy: the balance between the drum's tone and the snare rattle.

Timbre tone — the pitch balance and brightness of the two resonators.

Morph decay.

Aux the synthetic snare: harder, more processed, with more FM in the body.

Self-enveloped, same as the bass drum: Morph is the decay, and the LPG controls do nothing.

23 · Hi-Hat `HHAT`

Six square oscillators at inharmonic ratios plus noise, filtered and gated — the 808 recipe on the main output, with a more metallic alternative on the aux.

Harmonics noisiness: the balance between the square cluster and plain noise.

Timbre tone — the filtering, from dark and closed to bright and open.

Morph decay, which is what makes this the same model for both a closed and an open hat.

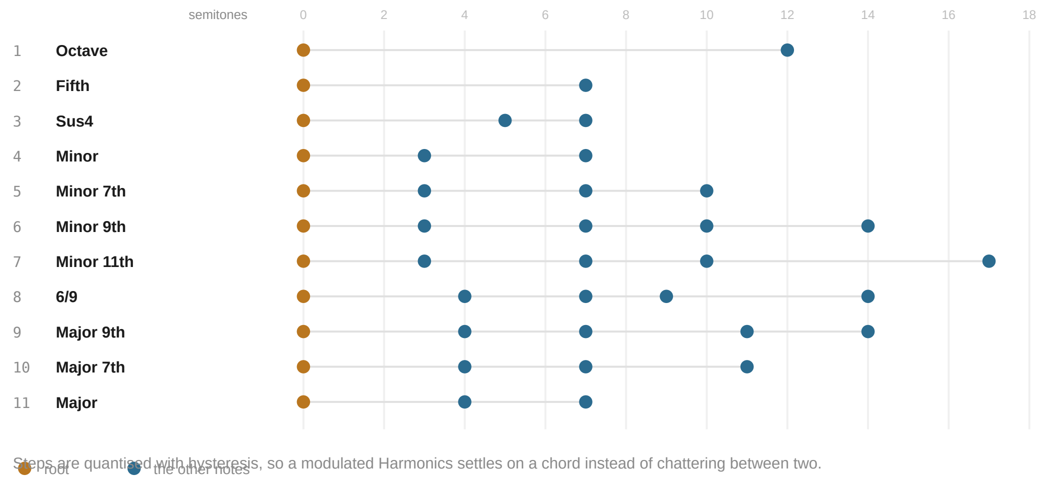
Aux the metallic variant: a different noise source, harder and more ringing.

Self-enveloped: Morph is the decay, the LPG controls are inert.

08 Quick reference

THE CHORD BANK, LOW TO HIGH

Harmonics steps through these eleven in Chords, String Machine and Chiptune.
The order runs from the emptiest intervals up through the minor family to the major one.



Steps are quantised with hysteresis, so a modulated Harmonics settles on a chord instead of chattering between two.

Figure 3.6 — The eleven chords Harmonics steps through, drawn as intervals. Sweeping it walks the harmony rather than jumping around it.

THE 24 MODELS, FOUR CHARACTERS EACH

In index order. A saved sound stores the raw index, so this order is fixed.

MODL is also what the Bouls model menu item calls itself, and HARM is also the Harmonics control. Four characters do not go far.

Figure 3.7 — The 24 models as four-character codes, in the index order a saved sound stores.

The three controls, twenty-four ways

#	MODEL	HARMONICS	TIMBRE	MORPH	AUX OUTPUT

0	Virtual Analog VCF	resonance / slope	cutoff	waveform + sub	high-pass output
1	Phase Distortion	modulator ratio	distortion amount	asymmetry	free-running (PM)
2–4	6-op FM A/B/C	patch (32)	brightness	envelope length	same as main
5	Wave Terrain	terrain (8)	orbit radius	orbit offset	sine variant
6	String Machine	chord	cutoff + ensemble	registration	other stereo side
7	Chiptune	chord	inversion / arp*	pulse width	bass note
8	Virtual Analog	detune	square PW / sync	saw shape	monster sync
9	Waveshaping	shaper curve	fold amount	slope / PW	sine + fold mix
10	2-op FM	ratio	FM index	feedback (bipolar)	sub-oscillator
11	Granular Formant	formant ratio	formant freq	window shape	second formant osc
12	Harmonic	peak count	peak position	slope	organ partials
13	Wavetable	bank + crush	row position	column position	5-bit copy
14	Chords	chord	inversion	registration → wavetable	root note
15	Speech	synth / word bank	formant shift	phoneme / word	excitation
16	Granular Cloud	spread	density	grain duration	sine swarm
17	Filtered Noise	filter mode	noise clock	resonance	dual band-pass
18	Particle Noise	frequency spread	density	diffusion / resonance	raw pulses

19	Inharmonic String	inharmonicity	pluck brightness	**decay**	exciter
20	Modal Resonator	inharmonicity	strike brightness	**decay**	exciter
21	Bass Drum	punch / drive	tone	**decay**	synthetic drum
22	Snare Drum	snappy	tone	**decay**	synthetic snare
23	Hi-Hat	noisiness	tone	**decay**	metallic variant

* Timbre selects the arpeggiator pattern instead of the inversion when the low-pass gate is on. Bold Morph entries mark the five models where Morph *is* the decay control, because the LPG's own decay is bypassed.

The chord bank

Harmonics steps through the same eleven chords in Chords, String Machine and Chiptune, in this order:

#	CHORD	#	CHORD	#	CHORD
1	Octave	5	Minor 7th	9	Major 9th
2	Fifth	6	Minor 9th	10	Major 7th
3	Sus4	7	Minor 11th	11	Major
4	Minor	8	6/9		

The order runs from the emptiest intervals up through the minor family to the major one, so sweeping Harmonics walks the harmony rather than jumping around it. Steps are quantised with hysteresis, so a modulated Harmonics settles on a chord instead of chattering between two.

09 Reference

CONTROL	DEFAULT	NOTES
Bouls model	8 — Virtual Analog	saved as a raw index; switchable mid-note
Harmonics	0	Osc 1 Pulse Width · CC 23

Timbre	0	Osc 1 Wave Index · CC 25 · lower half of CC / automation inert
Morph	0	Osc 2 Pulse Width · CC 28
Low-pass gate	off	inert on the eight self-enveloped models
LPG decay / LPG colour	25 / 25	read only while the LPG is on
Aux output	off (main output)	free to switch; identical to main on the FM banks

SWITCHING IT OFF ENTIRELY

`SETTINGS → COMMUNITY FEATURES → Bouls Engine` (`BOUL`), on by default. Switching it off removes `BOULS` from the oscillator type list only. A sound already set to Bouls keeps playing as Bouls; the menu just has no row to land on until the feature is back on, and shows the first entry rather than an out-of-range index in the meantime.

PROVENANCE

Émilie Gillet, Mutable Instruments · MIT licensed. The 24-model list follows upstream's own registration order from `plaits/dsp/voice.cc` , and the per-model control meanings in sections 07 and 08 were read from the engines themselves rather than from the module's manual. Source: `src/deluge/dsp/plaits/` , `src/deluge/dsp/plaits_adapter.cpp` , `src/deluge/gui/menu_item/plaits/` , `src/deluge/gui/menu_item/osc/type.h` .

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SYNTHSTROM DELUGE · RUDEMENT FIRMWARE

The

Wisps

Operating manual for the six-mode texture engine

Émilie Gillet's Clouds, as a Deluge insert effect at the head of the FX chain. A granular processor with six playback modes and eleven shared controls, taken from the communitymaintained Parasites continuation rather than the stock firmware, which is where the Oliverb and Resonestor modes come from. The DSP is byte-identical to upstream; everything Deluge-specific lives in a thin adapter that resamples 44.1 kHz down to the engine's native 32 kHz and back.

01 What it is

One recording buffer, about a second of audio, and six different ways of reading back out of it. Wisps is a texture generator rather than a colouring effect — it does not add an identifiable character to a signal so much as take it apart and rebuild it.

Six modes share eleven controls, and **the controls mean different things in each mode**. That is upstream's design, not a shortcut taken here: the same knob is grain density in Granular, diffusion in Stretch, reverb decay in Oliverb and resonator feedback in Resonestor. Section 03 gives the per-mode meaning of every control, and it is the part of this manual worth keeping open.

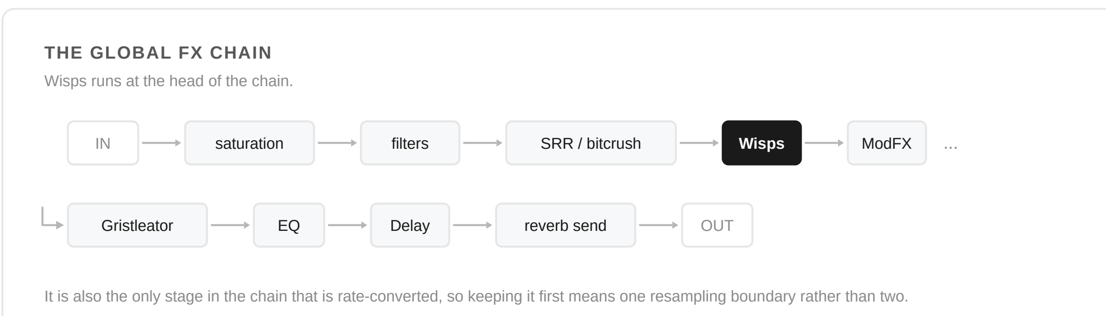

Figure 4.1 — Grain the raw signal and then shape the result, rather than graining something already drenched in delay — and one resampling boundary rather than two.

> **WHERE IT SITS IN THE CHAIN**
>
> Wisps runs at the head of the global FX chain — after the drums or clip render, saturation, the filters and the SRR/bitcrush distortion, but ahead of ModFX, the Gristleator, EQ, delay and the reverb send. The reasoning, from the implementation: it is a texture generator rather than a colouring effect, so the useful arrangement is to grain the raw signal and then shape the result, not to grain something already drenched in delay. It is also the only stage in the chain that is rate-converted, so keeping it first means one resampling boundary rather than two.

02 Finding it

FINDING WISPS

A global effect, so it appears wherever global FX live.

FX › **DUST**

Mode	DUST	Texture	Text
Blend	Blnd	Spread	Sprd
Position	Posn	Feedback	Fdbk
Size	Size	Reverb	Verb
Pitch	Ptch	Freeze	–
Density	Dens		

Song, Arranger, Automation, Kit affect-entire, Audio Clip and Performance View. Not on a synth clip or a single kit row — these are global parameters.

Figure 4.2 — The eleven controls, and the six places the menu appears.

Your first minute

#	DO THIS	AND WATCH
1	Open `FX → DUST` and set **Mode** to Granular	Nothing yet. The engine allocates its buffer, but Blend starts fully dry on purpose
2	Bring **Blend** up to about 30	Grains. This is the sound Wisps is known for
3	Take **Density** down to 28	Silence. That is upstream''s dead zone, not a fault — which is why the default sits at 40, clear of it
4	Put **Density** back and switch **Freeze** on	The buffer stops taking new audio and becomes an instrument. Sweep **Position** through what you captured
5	Change **Mode** to Oliverb	A click, and Blend drops back to dry. Six modes are six different processes, so one Blend position would mean six different amounts

Wisps is a global effect. It lives on the FX submenu wherever global FX live, and its 7-segment name is `DUST`.

CONTEXT	PATH
Song / Arranger / Automation view	`FX → DUST`
Kit, affect-entire (Kit Global FX)	`FX → DUST`
Audio Clip	`FX → DUST`

Performance View	`FX → DUST`
Synth clip, or a single kit row	not available — see below

> **WHY NOT ON SYNTH CLIPS**
>
> Wisps' eleven controls are global parameters. A synth voice carries a much shorter parameter set, and reading global parameters off a sound would run off the end of that array. The menu is therefore only ever hung off the global FX tree. There was an attempt to drive Wisps from a synth clip through a per-source send bus; it regressed the effect badly and was reverted. The one salvageable piece — a menu item that resolves Wisps' parameters to the song's parameter manager whatever context it was opened from — would give synth-clip control of a song-level Wisps without any routing work, and has not been re-applied.

> **AUTOMATION, MIDI AND THE SHORTCUT GRID**
>
> All nine continuous parameters are automatable and are saved with automation data. They are **not** in the Automation View parameter list, **not** on the shortcut grid, and have **no** default MIDI Follow assignments — but they can be MIDI-learned like any other parameter, and learned mappings persist in `SETTINGS/MIDIFollow.XML`. Mode and Freeze are not parameters at all, so neither can be automated.

03 The controls

Values below are the on-screen 0–50 reading. Only Pitch is bipolar; the other eight continuous controls run from bottom to top.

Mode default OFF

Which of the six engines is running: **Granular**, **Stretch**, **Delay**, **Spectral**, **Oliverb** or **Resonestor**. Granular, Stretch, Delay and Spectral are stock Clouds; Oliverb and Resonestor came with Parasites.

OFF is not a mute — it frees the engine's entire working buffer, and costs nothing at all. Changing mode is the one operation that clicks: the engine reinitialises fourteen sub-objects, and a 20 ms fade reduces but does not eliminate it.

Blend `Blnd` default 0 — fully dry, and back to dry on every mode change

Dry/wet, and it means the same thing in all six modes. It starts fully dry deliberately, so that selecting a mode reads as "nothing yet" rather than as an abrupt wall of grains — bring it up to hear the effect at all.

It returns to fully dry whenever the mode changes. The six modes are six different processes rather than six settings, so one Blend position means a different amount of signal in each of them; carried across a mode change it is a step of unpredictable size, landing on the engine you have not heard yet. Zeroing it makes that step always the same size and always downward — the new engine arrives silent and you bring it in. Re-confirming the mode already running leaves Blend where it is; browsing past a mode and back does not, because every detent is a real request.

Resonestor is the one mode upstream never crossfades — it spends its own dry/wet control on distortion instead. Here the crossfade is done after the engine, with upstream's equal-power curve so the taper matches the other five, and the distortion is held at zero.

Position `Posn` centre 25

Where in the recording buffer playback reads from — the "when", not the "what". In Delay it is the delay time; in Oliverb a pre-delay; in Spectral the magnitude store/replay position. In Resonestor it shapes an excitation burst that the Deluge never fires, so it does nothing there.

Size `Size` centre 25

Grain size in Granular — roughly 16 ms at the bottom to 512 ms at the top. Window size in Stretch, loop length in a frozen Delay, spectral warp in Spectral, room size in Oliverb, and *chord selection* in Resonestor.

Pitch `Ptch` bipolar · centre 25 = unity

Transposition, ±24 semitones — two octaves either side of unity, centre-detented like master pitch. Upstream clamps to ±48 internally; the narrower range is what Clouds' own front panel covers and keeps the granular player's resampling ratio sane.

In Delay and Oliverb the pitch shifter has a dead zone by design: below about 0.3 semitones of detune the shifted signal is silent, ramping to full by 0.7 semitones. Small detunes do nothing on purpose.

Density `Dens` default 40 — deliberately off-centre

In Granular, grain rate — and it is a meta-control: below the centre grains are triggered regularly, above they are randomised, and **at the centre there are no grains at all**. Upstream's dead zone is audibly silent from roughly menu 26 to 30. The default sits at 40, clear of it. *If Wisps ever appears silent, check Density before anything else.*

In every other mode it means something else entirely: diffusion in Stretch and Delay, refresh rate and phase randomisation in Spectral, **reverb decay** in Oliverb, and resonator feedback in Resonestor — where the law is so steep that essentially all the action is in the top of the travel.

Texture `Text` centre 25

In Granular, grain window shape — rectangular through gaussian for the first three quarters of the travel, then the remainder becomes diffusion. In Stretch and Delay it is a combined lowpass/highpass tone control sweeping through centre. Spectral quantisation in Spectral, tone in Oliverb, damping and narrowness in Resonestor.

Spread `Sprd` centre 25

Stereo behaviour, and it differs sharply by mode: per-grain random panning in Granular, a left/right swap-crossfade in Stretch and Delay, diffusion in Oliverb, stereo width and separation in Resonestor. It does nothing at all in Spectral.

Feedback `Fdbk` default 0 — off

Regeneration back into the input, with a highpass in the loop, in Granular, Stretch, Delay and Spectral. In Oliverb it is the modulation *rate* and in Resonestor the harmonicity — in neither of those is it feedback in any sense.

Reverb `Verb` default 0 — off

A reverb after the engine, in Granular, Stretch, Delay and Spectral. In Oliverb it is the modulation *amount*; in Resonestor the spread of the resonator taps. Neither is a reverb.

Freeze default off · latched

Holds the recording buffer: what is already in there keeps playing, but nothing new is written in. Upstream exposes this as a momentary button; here it is a latch, because the Deluge has no spare panel button for it and a latched freeze is the more useful of the two live.

Per mode: Granular, Stretch and Delay stop recording, and Delay switches to loop playback with Size as loop length and Pitch as playback speed. Spectral stops capturing magnitudes. Oliverb becomes an infinite hold with the input muted — and is the one mode that keeps recording while frozen. Freeze is hidden while Mode is OFF, where it would do nothing audible.

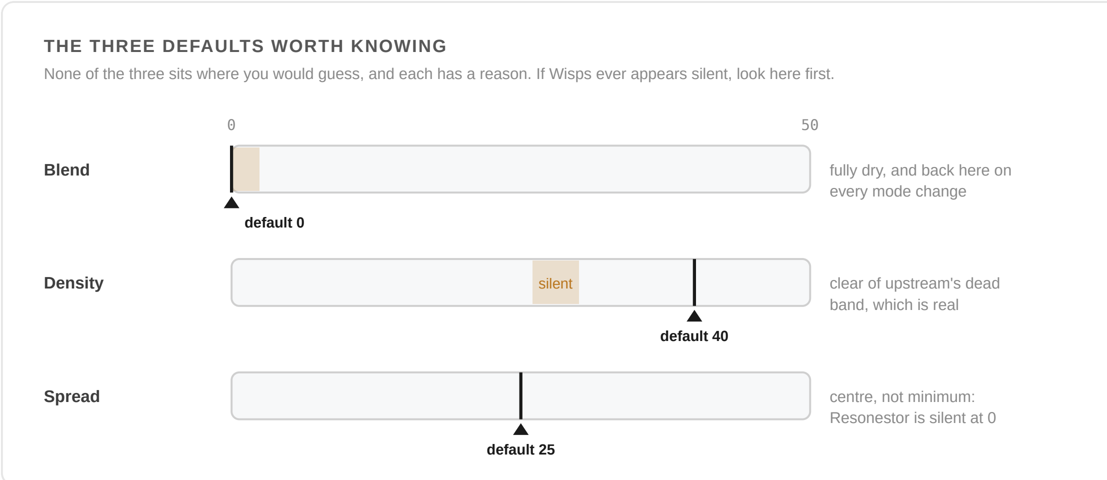

Figure 4.3 — Not one of the three sits where you would guess. Blend starts fully dry and is put back there on every mode change; Density's default clears a silent band that is upstream's, not a fault; and Spread sits at centre because Resonestor's tap reads are multiplied by zero at the minimum.

WHAT FREEZE DOES, MODE BY MODE

Upstream exposes this as a momentary button. Here it is a latch, because the Deluge has no spare panel button, and a latch is the more useful of the two live.

MODE	THE BUFFER	WHAT YOU HEAR
Granular	● stops recording	what is in the buffer keeps granulating
Stretch	● stops recording	the window keeps running over held audio
Delay	● stops recording	switches to loop playback — Size is loop length, Pitch is playback speed
Spectral	● stops capturing	magnitudes are held and replayed
Oliverb	● keeps recording	infinite hold with the input muted — the one mode that does not stop
Resonestor	● no effect	there is no recording buffer to hold

Freeze is hidden while Mode is OFF, where it would do nothing audible.

Figure 4.4 — Freeze is six different behaviours, one per mode — and Oliverb is the one that keeps recording.

The same eleven knobs, six ways

	GRANULAR	STRETCH	DELAY	SPECTRAL	OLIVERB	RESONESTOR
Position	read position	playhead	delay time	store/replay pos	pre-delay	inert
Size	grain size	window size	loop length*	spectral warp	room size	chord
Pitch	grain transpose	transpose	shift / speed*	shift ratio	ratio + decay	pitch

Density	grain rate	diffusion	diffusion	refresh + phase	decay	feedback
Texture	window shape	tone	tone	quantisation	tone	damping
Spread	grain pan	L/R swap	L/R swap	nothing	diffusion	width / sep.
Feedback	feedback	feedback	feedback	feedback	mod rate	harmonicity
Reverb	reverb	reverb	reverb	reverb	mod amount	tap spread
Blend	dry/wet	dry/wet	dry/wet	dry/wet	dry/wet	dry/wet

* when frozen.

04 Starting points

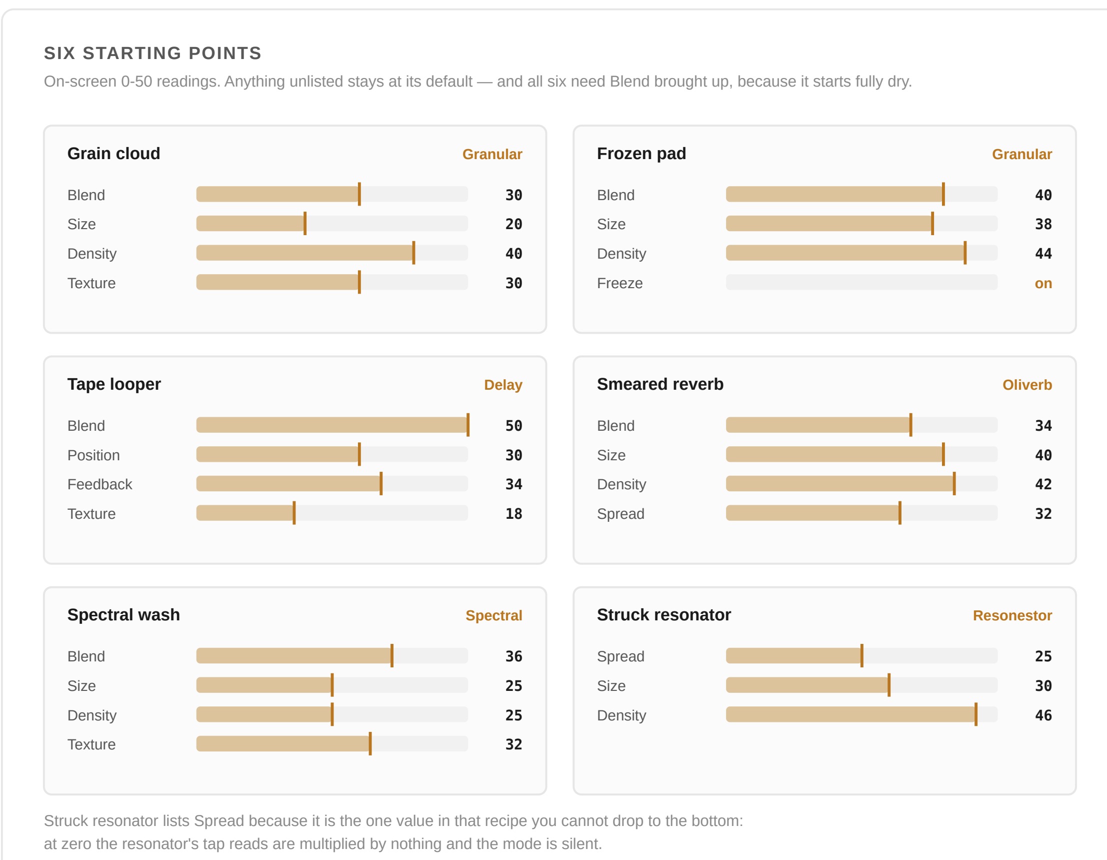

Figure 4.5 — Six settings worth dialling in as-is, then pulling apart.

Six settings worth dialling in as-is, then pulling apart. Values are on-screen 0–50 readings; anything unlisted stays at its default. All of them need Blend brought up — it starts fully dry.

Grain cloud Granular · Blend 30 · Size 20 · Density 40 · Texture 30

The sound Wisps is known for. Short grains, randomised triggering, window shape softened. Take Density up for a denser cloud, or below 26 for silence — which is the dead zone, not a fault.

Frozen pad Granular · Blend 40 · Size 38 · Density 44 · Freeze on

Play a chord, switch Freeze on, and the buffer becomes an instrument you can keep grinding after the source has stopped. Position sweeps through what you captured.

Tape looper Delay · Blend 50 · Position 30 · Feedback 34 · Texture 18

Position is delay time here, up to about a second. Texture below centre rolls the top off each repeat, so the loop degrades the way tape does.

Smeared reverb Oliverb · Blend 34 · Size 40 · Density 42 · Spread 32

Density is decay, Spread is diffusion, Feedback and Reverb are the modulation rate and depth. Add a semitone or two of Pitch for the shimmer.

Spectral wash Spectral · Blend 36 · Size 25 · Density 25 · Texture 32

The most expensive mode — roughly 2.4× Granular. Density at centre keeps phase coherent; move it either way to randomise. Spread does nothing here.

Struck resonator Resonestor · Spread 25 · Size 30 · Density 46

Spread has to stay off its minimum or this mode is silent — at zero the resonator's tap reads are multiplied by nothing. The default of 25 already clears it; the value is listed here because it is the one setting in this recipe you cannot drop to the bottom. Density is the resonator's decay and lives in the top few steps.

05 Reference

Defaults and displays

THE WISPS CONTROLS ON A 7-SEGMENT DISPLAY

Mode shows DUST; the nine continuous controls take four characters each.

Figure 4.6 — Mode shows DUST; the nine continuous controls take four characters each.

CONTROL	7SEG	DEFAULT	SAVED AS	INACTIVE AT
Mode	`DUST`	Off	`cloudsMode`	Off — nothing allocated
Blend	`Blnd`	0	`cloudsBlend`	0 — fully dry
Position	`Posn`	25	`cloudsPosition`	—
Size	`Size`	25	`cloudsSize`	—
Pitch	`Ptch`	25	`cloudsPitch`	25 — unity
Density	`Dens`	40	`cloudsDensity`	— (26–30 is silent)
Texture	`Text`	25	`cloudsTexture`	—
Spread	`Sprd`	25	`cloudsSpread`	—
Feedback	`Fdbk`	0	`cloudsFeedback`	0
Reverb	`Verb`	0	`cloudsReverb`	0
Freeze	`Freeze`	off	`cloudsFreeze`	off

Cost by mode

Host-relative measurements against the engine compiled natively through the adapter's exact plumbing. Ratios transfer to hardware; absolute figures do not, and the fraction of the Deluge's budget one instance eats has not been measured on device.

MODE	GRANULAR	DELAY	RESONESTOR	STRETCH	OLIVERB	SPECTRAL

Relative cost	1.0×	1.29×	1.84×	2.02×	2.15×	2.42×

The resampler alone is 8–19% of the stage depending on mode — not trivial beside the engine. Mode choice matters more than anything else here.

Rate, and what that costs

Wisps runs natively at 32 kHz in 16-bit stereo. The Deluge runs at 44.1 kHz. The adapter down-samples in, runs the engine in whole 32-frame blocks, and up-samples back out, using linear interpolation. That round trip costs about 44 samples — roughly 1 ms — and it is clean in the bass and low mids but poor at the top:

FREQUENCY	100 HZ	500 HZ	1 KHZ	2 KHZ	5 KHZ	10 KHZ	14 KHZ
Round-trip SNR	60 dB	46 dB	39 dB	31 dB	17 dB	7 dB	3 dB

Note also that the engine's own 32 kHz rate puts Nyquist at 16 kHz regardless of resampler quality. Wisps is not the stage to reach for when you want the top octave intact.

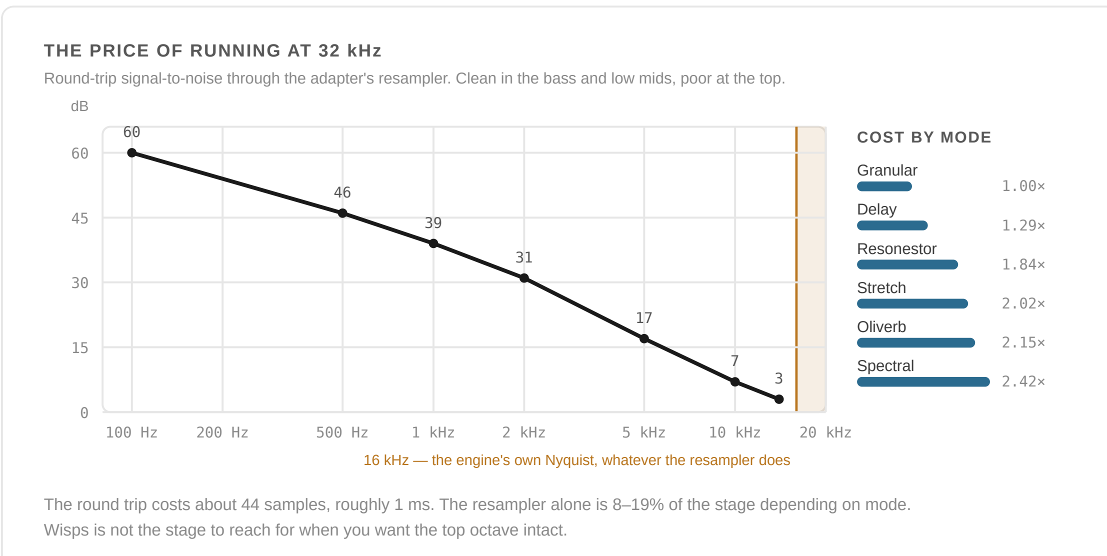

Figure 4.7 — Two costs in one picture: what the resampler does to the top of the band, and what each mode costs relative to Granular. Mode choice matters more than anything else here.

Memory

The engine needs one 180 KB working buffer. Nothing is allocated while Mode is `OFF`; the adapter appears the first time you leave OFF, and the buffer is taken on the first render after that. Selecting OFF again frees the whole thing rather than merely muting it. If the allocation fails you get an *insufficient RAM* error and the mode reverts to OFF; if the buffer is stolen by the sample cache the recording is lost but audio passes through dry rather than dropping.

Notes worth keeping in mind

- **Nothing is allocated until you leave OFF.** Wisps costs literally nothing while switched off, and selecting OFF gives the 180 KB straight back.
- **Old songs are safe.** The parameters were appended to the end of the global set rather than inserted, so no existing automation was repointed, and a song with no Wisps attribute loads with the mode Off.
- **Settings survive even when the mode is off.** All nine parameters are written unconditionally, so dialling in a texture, switching Wisps off to hear the dry signal, and saving does not lose them. Blend is the one exception, and only across a mode *change*: switching off to OFF and back is itself a mode change, so it returns dry.
- **The modes are levelled against each other.** Upstream's six are nowhere near level — measured full wet, Oliverb sits 9.9 dB above Granular and Resonestor 3.5 dB above it. Both are trimmed back to Granular on the wet path. Nothing is boosted: the quiet modes stay quiet, because their softness is upstream's voice rather than a fault.
- **Stereo throughout.** Two channels in, two out, each with its own recording buffer of about a second — and each instance carries its own resampler state, so two tracks running Wisps never cross-feed.
- **Some upstream features are not wired up.** The trigger and gate inputs are never driven, which makes Delay's tap-tempo sync unreachable, Resonestor's excitation burst never fire, and Spectral's glitch never run. The granular sub-parameters upstream exposes beyond the nine panel controls are left at their defaults.

> KNOWN LIMITATIONS
>
> **Density has a silent band** around menu 26–30 in Granular. This is upstream's behaviour. The default of 40 is chosen to sit clear of it.
>
> **Mode changes click.** Reduced by a fade, never eliminated.
>
> **Stretch can drop out** and recover. Only Stretch, and it costs about twice Granular — it may simply be too expensive. Not diagnosed to a root cause.
>
> **Re-verify against your own build.** The six modes were confirmed on hardware, then the line went through the Clouds merge and several rounds of fixes — the most recent of which changed how every engine arrives (dry, and levelled against each other). Behaviour described here is read from that source; treat anything you have not heard on your own unit as worth checking.

> SWITCHING IT OFF ENTIRELY
>
> `SETTINGS → COMMUNITY FEATURES → Wisps FX` (`DUST`), on by default. Switching it off hides every Wisps menu but leaves the processing alone, so a song already using Wisps continues to play exactly as saved.

PROVENANCE

Émilie Gillet, Mutable Instruments · the `mqtthiqs/parasites` community continuation, which is where Oliverb and Resonestor come from · MIT licensed · vendored August 2026. The DSP under `src/deluge/dsp/clouds/` is Gillet's original, unmodified — deliberately so, which is why the sample rate is adapted around rather than retuned in place: Clouds scatters `32000.0f` as a literal across several files rather than centralising it. Deluge-specific code lives only in `clouds_adapter.{h,cpp}`. This manual describes behaviour as built, and its figures are those stated in the implementation and its measurements.

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SYNTHSTROM DELUGE · RUDEMENT FIRMWARE

The

Gristleator

Operating manual for the ten-knob chop/throb effect

A modulation and distortion effect with a master switch, so it can be bypassed without unwinding its settings. After the Roy Gwinn design published in *Practical Electronics* in 1975 and sold in kit form by Phonosonics in 1977, which Chris Carter modified for Throbbing Gristle — the modification being that he brought the Bias trimpot out to the front panel, which is why Bias is a first-class knob here. The Eurorack redesign (FSS TG3/TG4, Gwinn + Carter, 2017) added Register and Resonance to the filter and Dirt to the VCA. This implements all nine, plus a master switch the hardware never had and never needed — a rack unit is bypassed by not patching it.

On · Rate · Depth · Shape · Bias · Mode · Freq · Reso · Dirt · Level

Edition of August 2026

01 What it is

One LFO driving two parallel stages — an amplitude stage (the "chop") and a state-variable filter (the "throb") — blended against each other by Mode. Ten controls in total: the master switch plus nine that describe the sound. There is no delay line; the whole effect costs six words of state per instance.

The Gristleator sits late in the global FX chain.

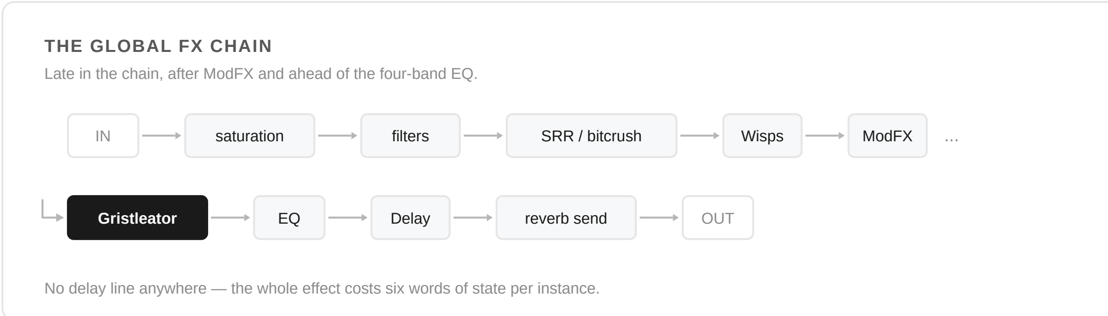


Figure 5.1 — Where the Gristleator sits: after ModFX, ahead of the EQ, and well before the delay and reverb send.

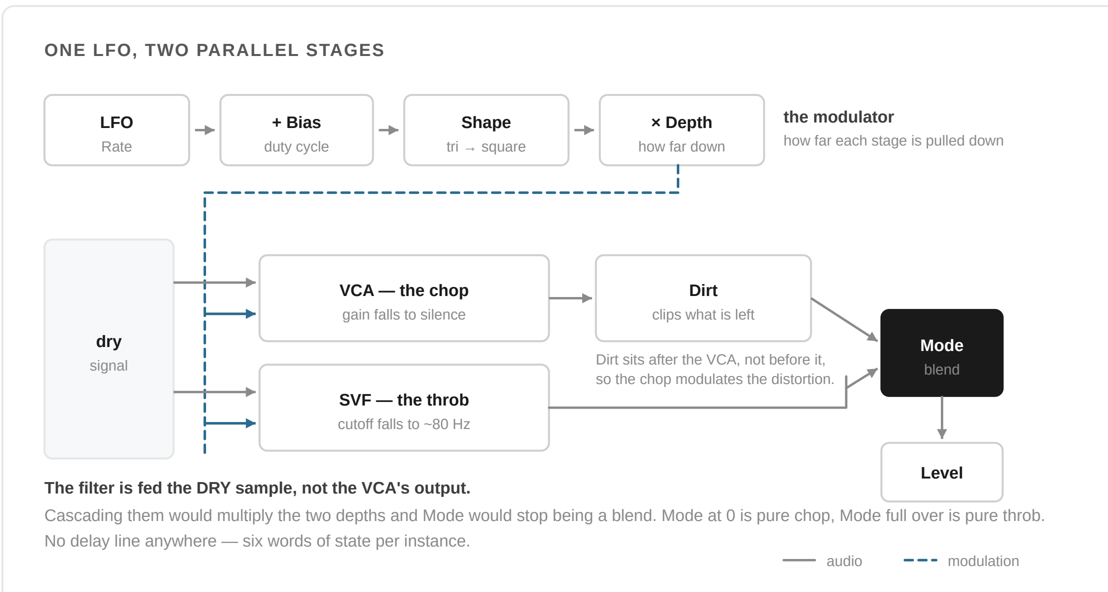


Figure 5.2 — Ten controls, and where each of them acts. The LFO is shaped once and drives both stages; Mode decides how much of each you hear.

> **WHY THIS IS NOT A MODFX TYPE**
>
> It used to be one, sharing the mod-FX Feedback and Offset params for Shape and Mode. But the global effect chain silences mod FX automatically whenever the currently selected gold-knob parameter sits at its minimum, and that parameter defaults to Feedback — which is where Shape lived. Shape at minimum is a perfectly musical setting (a plain triangle wave, and where an untouched knob starts), so on song, kit and audio-clip FX the whole effect silently vanished the moment nobody had touched it. Owning its own ten params instead of borrowing mod-FX's four removes that gate entirely: every knob here has a musically valid minimum, and none of them can make the effect disappear on their own.

02 Finding it

Your first minute

#	DO THIS	AND WATCH
1	Open `FX → GRIS`	**On** is the first item, because it is the only control that can make the other nine inaudible
2	Switch **On** on, then take **Depth** to about 35	Tremolo. Most of the audible change is in the bottom half of that knob
3	Take **Shape** to maximum	The triangle squares off and the chop reads as a hard gate rather than a swell
4	Push **Mode** fully over	The amplitude chop disappears and the LFO sweeps a resonant filter instead. Turning Depth down here does *not* bypass it — a static filter stays in the path
5	Take **Rate** to the very top	The LFO crosses into audio rate and the chop stops being tremolo and starts being a second tone

Ten shared unpatched params, so one set of menu items serves synths, kits, audio clips and song FX alike — unlike Sear, which is per-voice and therefore synth-clip only. The menu paginates four items per page (4/4/2 across three pages). **On is first** deliberately: it is the only control that can make the other nine inaudible, so it has to be reachable without paging past it.

03 The master switch

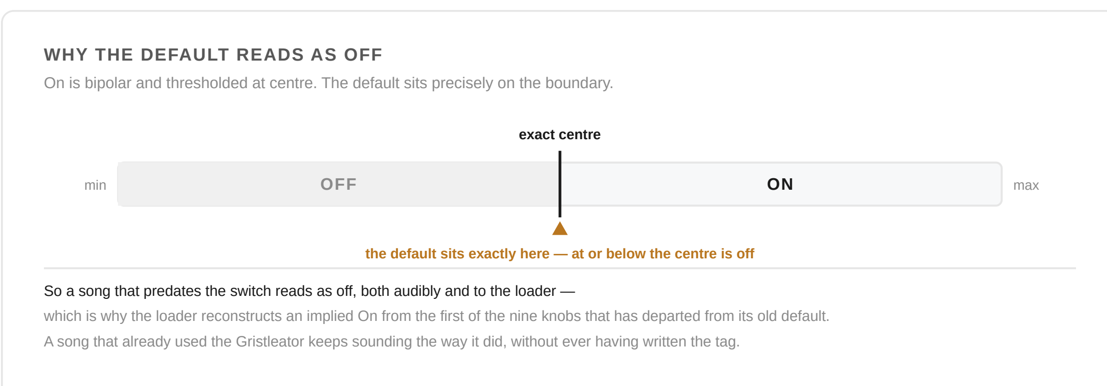


Figure 5.3 — On is bipolar and thresholded at centre, and the default sits precisely on the boundary — which is what makes it read as off to a song that predates the switch.

On is bipolar and thresholded at centre — anything above the exact centre point is on, at or below it is off. The default sits precisely on that boundary, which reads as off both audibly and to a song that predates the switch.

> **WHAT THIS REPLACED**
>
> Before the switch existed, "on" was inferred from whether any of the four audibly active knobs — Depth, Mode, Dirt, Level — sat away from its default. That was correct in a narrow sense, but it made the effect impossible to bypass without also discarding the patch: every one of the nine knobs has a musically valid setting that isn't its default, so "return everything to default" was the only off switch, and it wasn't reversible. It was also backwards in one specific way that tripped people up repeatedly — turning Depth down to zero looks like an off switch and isn't one, because Mode pushed toward the filter leaves a static lowpass sitting in the signal path at whatever Freq is set to, exactly as the original hardware's VCF doesn't leave the circuit just because the LFO is turned down. The switch removes both problems: the nine knobs only ever describe the sound now, and one param decides whether you hear it.
>
> Songs saved before the switch existed carry the nine param tags and no "on" tag. The loader treats the first tag that departs from its old default as proof the effect was in use, and reconstructs an implied "on" from it — so a song that already used the Gristleator keeps sounding the way it did without ever having written the tag itself.

04 The controls

In signal order: the driver, then what it drives, then the output stage.

CONTROL	DEFAULT	WHAT IT DOES
Rate	centre	LFO speed. Rates through the knob's whole travel, from a slow sweep up into audio rate — pushed far enough, the chop stops reading as tremolo and starts reading as a second tone.
Depth	0 (min)	How far the LFO pulls the stage down. Zero is an exact bypass of the modulation — both stages still pass signal, just unmodulated. Warped so roughly −6 dB of chop already arrives by mid-travel; most of the audible change is in the bottom half of the knob, not the top.
Shape	0 (min)	Triangle → trapezoid → square, continuously. Minimum is a plain triangle.
Bias	centre	Offsets the LFO before shaping. Bipolar; pushed hard enough it rails the LFO fully open or fully shut — faithful to Carter's front-panel mod, not a bug.

Mode	0 (min — VCA)	0 = pure VCA (chop), full = pure VCF (throb), continuously blendable in between.
Freq ("Register")	centre (~800 Hz)	Filter centre. Bipolar, ±2 octaves, so the knob covers roughly 200 Hz to 3.2 kHz.
Reso	0 (min)	Filter resonance. Stops short of self-oscillation by design — a saturating stage in the filter's feedback path bounds it, so the top of the knob is a resonant sweep, not a squeal.
Dirt	0 (min — clean)	Blends the clean VCA output against a driven, soft-clipped one. Drive rises alongside the mix — up to 3× — so turning Dirt up both adds distortion and pushes it harder, the way a JFET stage in the original circuit distorts more as it's driven harder. Sits after the VCA, so it's the chop itself being distorted, not a static overdrive with a tremolo bolted on after it.
Level	max (unity)	Output trim.

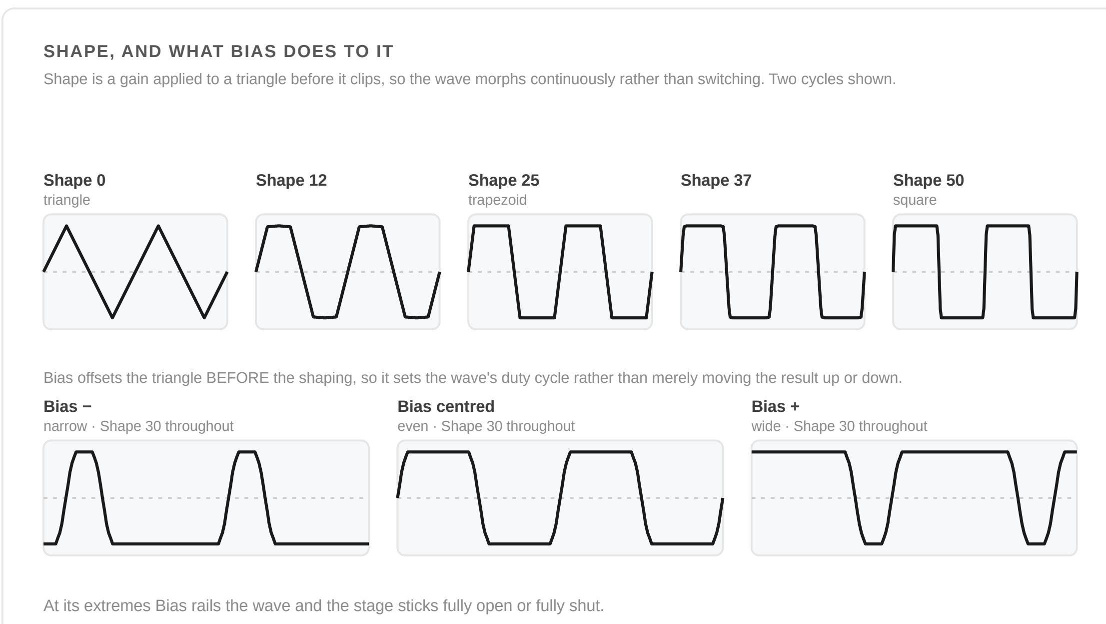


Figure 5.4 — Shape and Bias, computed through the shipped shaping code. Bias is added to the triangle before the shaping gain, which is why it sets the duty cycle rather than just offsetting the result.

05 Starting points

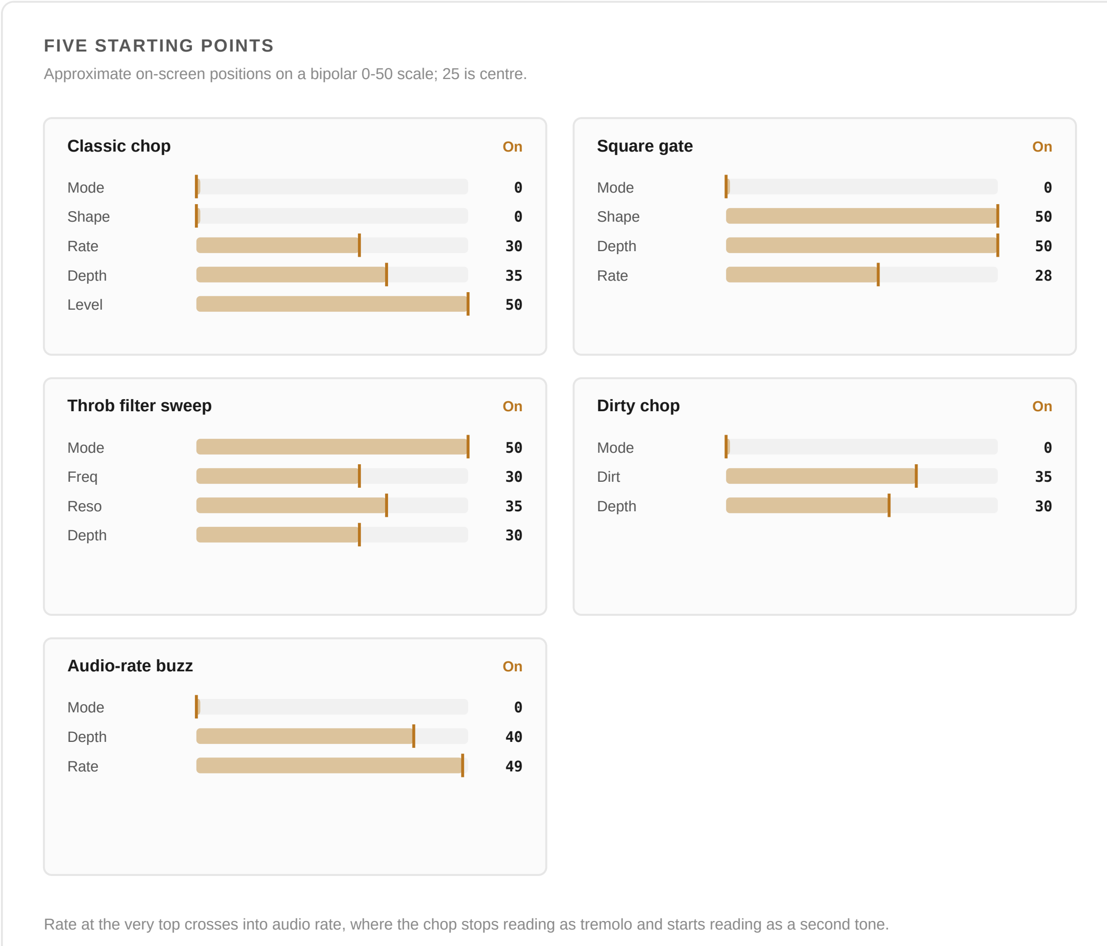


Figure 5.5 — Five settings worth dialling in as-is, then pulling apart.

Five settings worth dialling in as-is, then pulling apart. Knob values are approximate on-screen positions on a bipolar 0–50 scale (25 = centre).

Classic chop On · Mode 0 · Shape 0 · Rate 30 · Depth 35 · Level 50

Steady tremolo — a plain triangle LFO chopping the amplitude at a moderate rate.

Square gate On · Mode 0 · Shape 50 · Depth 50 · Rate 28

Shape at maximum turns the LFO into a near-square wave, so the chop reads as a hard gate rather than a swell.

Throb filter sweep On · Mode 50 · Freq 30 · Reso 35 · Depth 30

Mode fully toward the filter removes the amplitude chop entirely — the LFO sweeps a resonant filter instead, with no gating.

Dirty chop On · Mode 0 · Dirt 35 · Depth 30

The chop itself gets driven harder as it opens, rather than a clean chop followed by static distortion.

Audio-rate buzz On · Mode 0 · Depth 40 · Rate 48–50

Push Rate to the very top of its travel and the LFO crosses into audio-rate territory — the chop stops sounding like tremolo and starts sounding like a second, ring-mod-like tone.

06 Reference

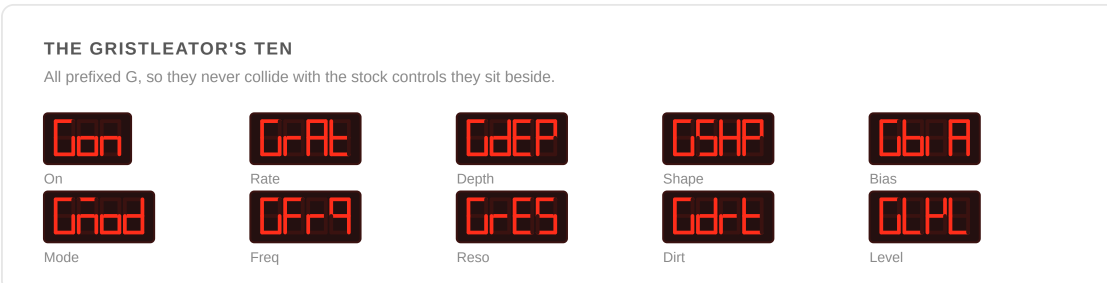


Figure 5.6 — All ten, prefixed G so they never collide with the stock controls beside them.

CONTROL	7SEG	SAVED AS
On	`GON`	`gristleOn`
Rate	`GRAT`	`gristleRate`
Depth	`GDEP`	`gristleDepth`
Shape	`GSHP`	`gristleShape`
Bias	`GBIA`	`gristleBias`
Mode	`GMOD`	`gristleMode`
Freq	`GFRQ`	`gristleFreq`
Reso	`GRES`	`gristleRes`
Dirt	`GDRT`	`gristleDirt`
Level	`GLVL`	`gristleLevel`

KNOWN LIMITATIONS

Bypass behaviour against the LFO timeline is not yet confirmed on hardware. This is open as issue #2 in the fork's tracker: whether switching On off and back on resumes the LFO phase cleanly or restarts it. Worth checking on real hardware rather than assuming either way.

SWITCHING IT OFF ENTIRELY

`SETTINGS → COMMUNITY FEATURES → Gristleator` (`GRIS`), on by default. Hides the Gristleator menu only; a song already using it keeps sounding exactly as saved.

PROVENANCE

Roy Gwinn, *Practical Electronics*, 1975 · Phonosonics kit, 1977 · Chris Carter's Throbbing Gristle front-panel modification · FSS TG3/TG4 Eurorack reissue, Gwinn + Carter, 2017. The soft clipper in the Dirt stage (`softClipCubic`) is shared with Sear. Source: `src/deluge/dsp/gristle.hpp` .

↑ Back to contents

SYNTHSTROM DELUGE · RUDEMENT FIRMWARE

The

Sear

Operating manual for the auto-levelling drive stage

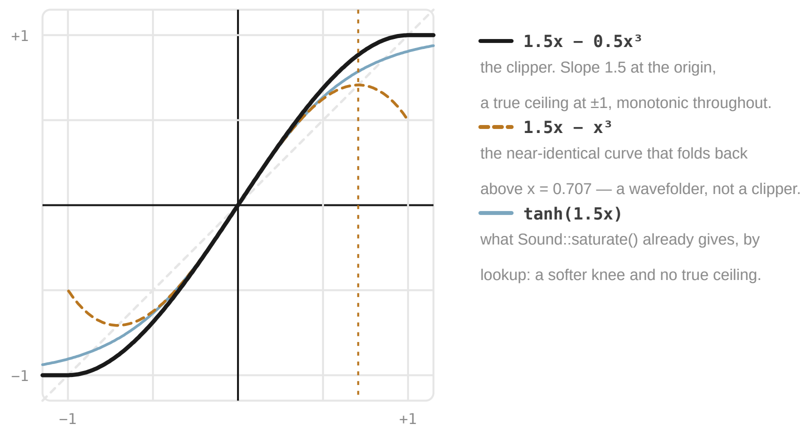
A cubic soft-clipper with automatic level matching, followed by a one-pole tilt tone control. Per-voice and patched, so it behaves like any other modulation destination. Was called Heat until 2026-08-08.

01 What it is

A cubic transfer curve rather than a second tanh — `Sound::saturate()` already gives a tanh-shaped drive via lookup table, so a second one would just duplicate an existing character. The cubic has a harder knee and a true ceiling at ±1, which reads as "pedal" beside the softer stock saturation, and it costs three multiplies with no table lookup, which matters running per voice.

THE CLIPPER, AND THE CURVE IT IS NOT

$y = 1.5x - 0.5x^3$, evaluated on [−1, 1].



Three multiplies, no lookup table — which matters when this runs per voice. There is no oversampling, so bright material driven hard will alias.

Figure 6.1 — The curve, and the two it was chosen over. The near-identical `1.5x − x³` is the reason the coefficient matters: it folds back above x = 0.707 and would have made this a second wavefolder.

ALIASING — A GENUINE LIMITATION, NOT AN OVERSIGHT

There is no oversampling, and the cubic curve generates harmonics up to 3× the input frequency, so bright material driven hard **will** alias. Oversampling per voice on the Deluge's processor was judged too expensive. Sear sits after the ladder filter in the signal path, so rolling the low-pass down also tames the aliasing — that's the practical fix, not a setting on the effect itself.

THE RENAME

This effect shipped as "Heat" until 2026-08-08, then was renamed Sear — XML attributes included (`sear` , `searTone`). That was safe only because the effect had never shipped and no saved song referenced the old tags. That window is closed: any future rename would need a read-side alias for the old spelling rather than a clean swap.

02 Finding it

Your first minute

#	DO THIS	AND WATCH
1	On a synth clip, open `FX → Distortion`	**Sear** and **STon** sit with the other per-voice distortion stages
2	Bring **Sear** up from the bottom	Audible by knob 3–5. There is no dead zone; the taper is a constant ratio per step
3	Take it to 25, then 50	25× pre-gain, then 406×. The output barely gets louder — the level matching is removing what the drive would otherwise add
4	Play softly, then hard	Your dynamics survive. It tracks the ratio across the clipper, not a level, so it is not behaving like a compressor
5	Take **STon** either side of centre	A tilt. Centre is a true bypass, measured flat, which is the whole point of the topology

`FX → Distortion`, alongside the other per-voice distortion stages — Saturation, Bitcrush, Decimation and Wavefold — as **Sear** (the drive) and **STon** (its tone control). Because it is patched and per-voice, it appears on synth clips and drums, not on kit-global or song FX.

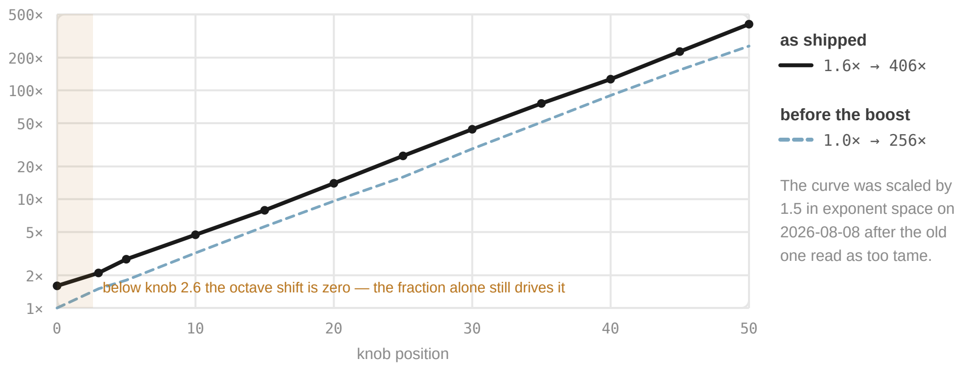
03 Sear (the drive)

A patched param, so it can be modulated like any other. Pre-gain sweeps on an exponential taper — roughly a constant ratio per knob step, audible from knob position 3–5 with no dead zone:

KNOB	0	3	5	10	15	20	25	30	35	40	45	50
Gain	1.6×	2.1×	2.8×	4.7×	7.9×	14×	25×	44×	76×	127×	228×	406×

THE DRIVE TAPER

Pre-gain into the clipper, measured through the real param pipeline. Log scale — a straight line is a constant ratio per step.

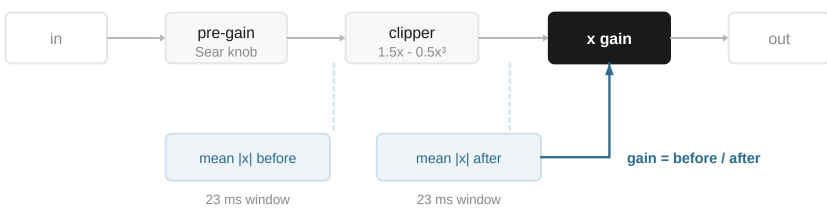


The auto-level stage removes the loudness the extra drive would otherwise add, so the boost lands as saturation rather than volume.

Figure 6.2 — Pre-gain into the clipper, on a log scale, so a straight line is a constant ratio per knob step. The dashed line is the taper as it shipped before 2026-08-08, kept here because the two are easy to confuse in older notes.

WHY THIS IS NOT A COMPRESSOR

Two followers, one either side of the clipper. The correction is their ratio — not a level.



A compressor tracks a level, so a quiet passage gets pushed up.

This tracks a ratio, so when the input gets quieter the output does too and the ratio barely moves — ordinary note dynamics pass through untouched.

Only the loudness the clipper is adding gets corrected, and a small lift is left in on purpose: about +2.3 dB at the top of the knob, so drive reads as drive.

Figure 6.3 — Two followers, one either side of the clipper, and the correction is their ratio. That is what makes this not a compressor.

AUTOMATIC LEVEL MATCHING

Two one-pole followers track mean absolute level either side of the clipper, over roughly a 23 ms window, and the correction applied is their ratio. **This is not a compressor and does not behave like one** — it tracks the ratio of two levels, not a level, so when the input gets quieter the output gets quieter with it and the ratio barely moves. Ordinary note dynamics pass through untouched; only the amount the clipper is *adding* is corrected. A small residual loudness lift is kept in on purpose — correcting all the way to a flat 0 dB made the drive read as "nothing is happening," so the target rises gently with knob position, measuring about +2.3 dB of real output rise at the top of the knob.

Cold starts are handled by seeding each new voice from the Sound's own running state rather than starting silent: only the very first note a Sound ever plays is measured cold, and every note after that inherits an already-warm gain estimate.

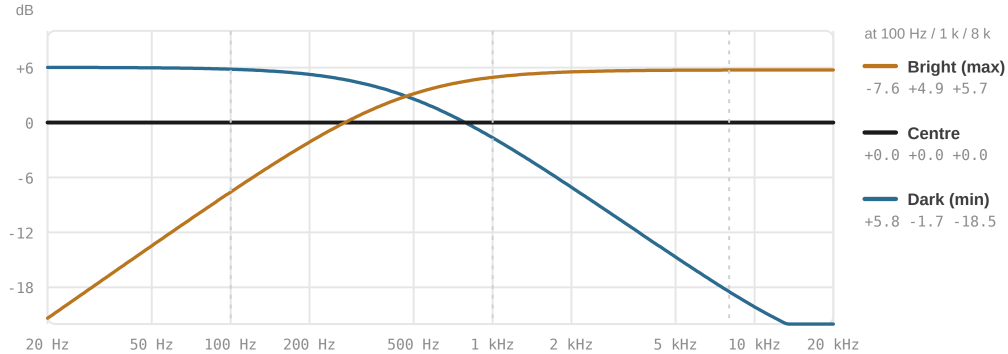
04 STon (the tone control)

An unpatched tilt tone control shaping what the drive bites on. A one-pole split into low and high halves, weighted so that at centre both halves sum back to the input exactly — measured flat to 0.00 dB. The pivot sits near 1 kHz, where a guitar-pedal tone stack usually puts it.

	100 HZ	1 KHZ	8 KHZ
Dark (min)	+5.8 dB	−1.7 dB	−18.5 dB
Centre (default)	0.0 dB	0.0 dB	0.0 dB
Bright (max)	−7.6 dB	+4.9 dB	+5.7 dB

STON — A TILT, NOT A FILTER

One lowpass split into two halves and weighted against each other. At centre the halves sum back to the input exactly.



The pivot sits near 1 kHz, where a guitar-pedal tone stack usually puts it. Centre measures flat to 0.00 dB.

Figure 6.4 — The tilt, computed from the shipped one-pole. The three measured points quoted in the table above are marked; centre is a true bypass, which is the whole point of the topology.

05 Reference

SEAR, AND ITS TONE CONTROL

Two controls, and only one of them has a default CC.

the drive · CC 22

the tilt · no default CC

Figure 6.5 — Two controls, and only one of them has a default CC.

CONTROL	7SEG	DEFAULT
Sear	`SEAR`	minimum (bypass)
STon	`STon`	centre (flat)

KNOWN LIMITATIONS

Aliasing on bright or hard-driven material — see above; there is no built-in remedy beyond rolling off the filter ahead of it. The per-block level-matching correction involves one 64-bit divide per voice when Sear is enabled; arithmetic puts that well under 1% of budget even at full polyphony, but it has not been profiled on hardware.

SWITCHING IT OFF ENTIRELY

`SETTINGS → COMMUNITY FEATURES → Sear` (`SEAR`), on by default. Hides the Sear drive and tone controls; processing already dialled in keeps running.

> **PROVENANCE**
>
> Originally shipped as "Heat"; renamed Sear 2026-08-08. Shares its soft clipper (`softClipCubic`) with the Gristleator's Dirt stage. Source: `src/deluge/dsp/sear.hpp` .

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The

Kit Split

Operating manual for exploding a kit into single-sound kits

Explodes a kit into one single-sound kit per drum that has notes, carrying every clip on that kit across with it. Unlike every other feature in this manual, Kit Split ships **off** by default, because it is destructive.

Kit Global FX · Actions · Split Kit

Edition of August 2026

01 What it is

Every drum that has notes becomes its own kit holding exactly that one sound, with its own clip carrying only that drum's notes. In grid mode they land as N new columns where the source track was, drum 0 leftmost. This is a **move**, not a copy — the source kit is consumed.

The unit is the kit, not the clip. If a kit carries a part and a variation of it — two clips in the same grid column, different sections — both are split in the same action, and each new kit receives one clip per source clip at its original section. Twelve drums with two variations gives twelve columns, each with the part on top and the variation underneath, exactly as they went in. It does not matter which of the clips you invoke it from.

This is not a convenience. Drums are moved rather than copied, so splitting one clip on its own would pull the drums out from under its siblings and leave them pointing at sounds that had moved to other kits. Taking the whole kit at once is the only coherent way to do it.

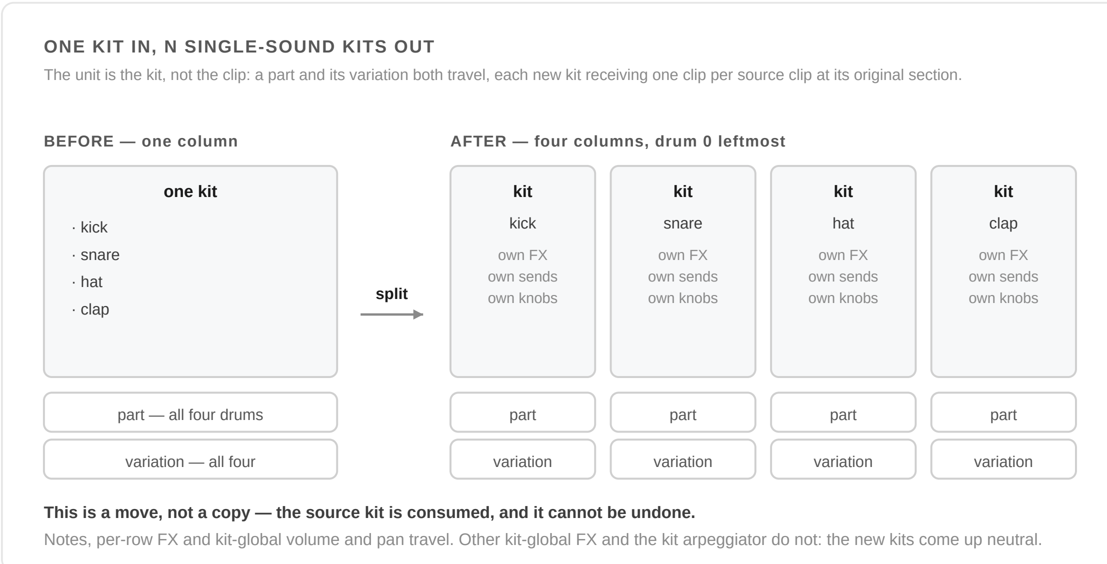

Figure 7.1 — Four drums and two clips in, four kits and eight clips out. Each new kit is a real track, with the global FX chain, sends and gold knobs a kit row never had.

02 What it's for

The plain reason is arrangement: a row you want to treat as its own track, with its own clip in the arranger, its own mute, and its own place in the grid.

The reason worth knowing about is **preparing a song for stem export.** Not because you need it to get separate files — the Deluge already exports a kit's drums as individual stems on its own. The reason is what a kit cannot do: every drum in one shares a single set of kit-global FX. One delay, one reverb, one filter, across all of them. Exporting drum stems gives you every drum through that same shared chain, or — with `Kit FX` switched off in `Configure Export` — every drum dry. What it will not give you is the kick through one treatment and the hats through another.

Splitting first changes that. Each sound becomes a real track with its own global FX chain, its own sends, its own gold knobs and its own automation, so you can treat each one differently on the Deluge and then export them as ordinary track stems — already sitting the way you want them before anything reaches a DAW.

> **SPLITTING FOR EXPORT? DO IT LAST**
>
> Kit-global FX do not travel — section 04. The new kits come up neutral, so whatever you had set on the kit bus is gone. Dial the per-sound treatments in *after* the split rather than before.
>
> And because the split is destructive and cannot be undone, save the song first. The usual order is: finish the arrangement, save, split, process each sound, export.

03 Using it

FINDING KIT SPLIT

On a kit clip, with affect-entire on.

`Kit Global FX` › `Actions` › **`Split Kit`** ships off — switch it on first

└ Split into N `SP12`

Stepping into the submenu is the confirmation. The label states how many kits you are about to get; on a 7-segment display the count survives as the last two digits.

Figure 7.2 — Where the action lives, and what confirms it.

Your first minute

#	DO THIS	AND WATCH
1	Switch the feature on at `SETTINGS → COMMUNITY FEATURES → Kit Split`	It is the one feature that ships off, because it consumes the kit it is given
2	**Save the song.**	This cannot be undone, and the undo history is cleared before anything moves
3	Mute the kit track	The rest of the song can keep playing. If anything on this kit is still sounding it will refuse with `MUTE`
4	`Kit Global FX → Actions → Split Kit`	The label states how many kits you are about to get. Stepping into it is the confirmation
5	Look at session view	N new columns where the source track was, drum 0 leftmost, each already in affect-entire with its gold knobs live — and all of them muted, waiting for you

On a kit clip, with affect-entire on:

```
Kit Global FX → Actions → Split Kit → "Split into N"
```

Stepping into the submenu is the confirmation — the action sits one level down on purpose, and its own label states how many kits you are about to get. On a 7SEG that label reads `SP12` rather than the full sentence, so the count survives the four digits. From there it just happens, landing you in session view with the new columns already in place, each new clip already in affect-entire with its one drum selected and its gold knobs live.

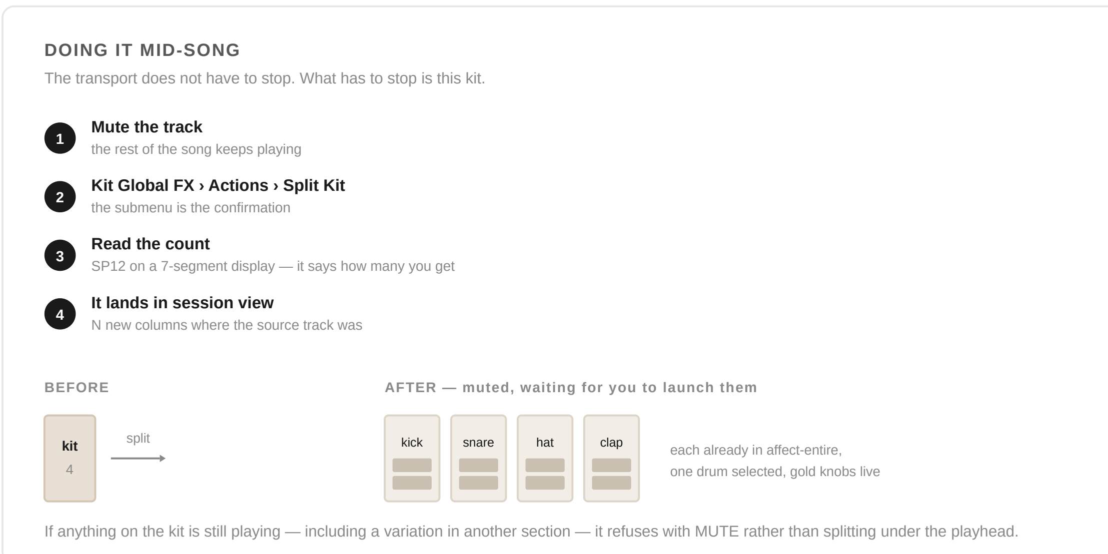

Figure 7.3 — Mute the track, split it, and launch the new ones when you want them. The transport never has to stop.

> **YOU CAN DO THIS MID-SONG**
>
> The transport does not have to stop. What has to stop is *this kit*: mute the track and split it while the rest of the song keeps playing, with no gap and no audible seam. The new tracks arrive muted, so you launch them when you want them — which also gives you a look at what you did before it reaches the mix.
>
> If anything on the kit is still playing — including a variation in another section that you did not invoke this from — it refuses with `Mute the track first` (`MUTE` on 7SEG) rather than splitting underneath the playhead.

04 What travels, and what doesn't

WHAT SURVIVES THE SPLIT

Each new kit holds exactly one drum, so anything that describes that one drum comes with it. Anything shared by the whole kit cannot.

TRAVELS

- ✓ Notes, velocity, probability, iterance, fill
- ✓ Per-row FX — filters, envelopes, patch cables, sends, sidechain
- ✓ Kit-global volume and pan
  values only, read per source clip
- ✓ Every other clip on the kit
  one clip per source clip, at its own section

DOES NOT

- ✗ Kit-global delay, reverb, filter, mod FX
  inheriting one delay would make N
- ✗ Kit arpeggiator
  it indexed drums that no longer share a kit
- ✗ Automation on volume or pan
  the values travel; the lanes do not
- ✗ Drums with no notes anywhere
  they go away with the source kit

Volume and pan travel because they are the output's gain and position rather than a wet effect — a kit at volume 43 comes back as N kits each at 43, not at the neutral default.

Figure 7.4 — Anything describing one drum comes with it; anything shared by the whole kit cannot.

	TRAVELS?
Notes, velocity, probability, iterance, fill	Yes
Per-row FX — filters, envelopes, patch cables, sends, sidechain	Yes
Kit-global volume and pan	Yes
Other kit-global FX — delay, reverb, filter, mod FX	**No** — new kits come up neutral
Kit arpeggiator	**No** — it indexed drums that no longer share a kit
Other clips on the same kit	Yes — each new kit gets one clip per source clip, at its own section
Drums with no notes anywhere on the kit	Skipped; they go away with the source kit

Volume and pan travel because they are the output's gain and position rather than a wet effect, and each new kit holds exactly one drum — copying them across reproduces the mix the source had rather than multiplying it. A kit sitting at volume 43 comes back as N kits each at 43, not at the neutral default. Other kit-global FX are destroyed deliberately: inheriting a delay would have turned one delay into N.

Values only — automation on the source's volume or pan is not carried across. Volume and pan are read per source clip, so a part and its variation keep their own levels. New kits are named from their drum, with a number appended on collision.

A drum counts as having notes if *any* clip on the kit has notes for it. A drum whose part exists only in the variation still gets its own kit — otherwise that part would be silently thrown away. Where a clip has nothing to say about a drum, that new kit simply gets one fewer clip rather than an empty one.

05 When it refuses

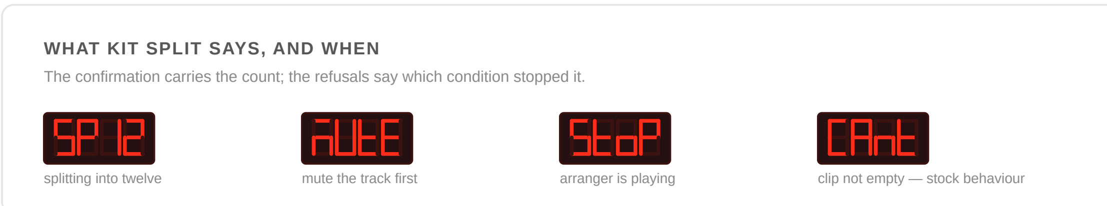

Figure 7.5 — The confirmation carries the count; each refusal names the condition that stopped it.

It will not run, and says so, if:

- any clip on the kit is playing — mute the track (`MUTE`)
- the arranger is playing, or you are recording to the arranger (`STOP`)
- fewer than two drums on the kit have notes
- any clip on the kit is the sync-scaling clip
- the kit has any instances in the arranger (splitting would gut them)
- more than 64 drums have notes, or the kit carries more than 16 clips

Several of these read wider than they used to, and deliberately so: the whole kit is being consumed, so a condition on any one of its clips is a condition on the operation.

> **NOT A BUG**
>
> Pressing `MIDI` or `CV` on a split track says *Clip not empty* (`CANT` on 7SEG). That's stock Deluge behaviour, not something the split introduced — any kit clip with notes refuses the same conversion.

NOT UNDOABLE

The undo history is cleared before anything moves, exactly as creating a new instrument does. Every kit and clip is built before anything is moved, so if RAM runs out the song is untouched and you get an error rather than half a split. If something fails partway through committing, the source clips stay in place holding whatever drums haven't moved yet — nothing is lost, but nothing rolls back either.

06 Reference

AVAILABLE ON BOTH FIRMWARE LINES

Kit Split ships as a small patch on top of each line (`feat/kit-split-13` , `feat/kit-split-muted-12`) rather than being merged into the day-to-day build. The behaviour described in this chapter is identical on both. They differ only where the APIs force it: drum naming (1.3 reads a name from MIDI/gate drums via `getDrumName()` ; 1.2.1 has no such method, so those kits come out as `SPLIT` , `SPLIT2` … instead), the instrument-creation call, the on-card flag, and where the menu hangs — 1.3 adds the split to an existing Kit Global FX → Actions menu; 1.2.1 has no such submenu, so this patch creates it.

Display strings work the same way on both: the entry goes in `english.json` and `seven_segment.json` , and the `g_*.cpp` tables are generated from those by `generate.py` at build time. Editing a generated table by hand appears to work and then silently stops working at the next clean build, when the generator rewrites it from the JSON and the string drops out of the table — the popup comes up blank rather than failing to compile.

SWITCHING IT ON

`SETTINGS → COMMUNITY FEATURES → Kit Split` (`SPLT`), off by default — the only feature in this fork that ships disabled.

PROVENANCE

Original to this fork, 2026.

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The

Aux Sends

Operating manual for the CV output send bus

The two CV sockets, used as an audio send bus. Turn a send up on a track and that track's audio appears at the socket — on its own, independent of the main mix — so it can feed a pedal, a mixer channel or a modular case while everything else carries on out of the mains.

MAIN · CV1 Send · CV2 Send · CV1/2 Level · Stereo Split

Edition of August 2026

01 What it is

> **NOT IN EVERY BUILD — CHECK BEFORE YOU GO LOOKING**
>
> Aux Sends lives on the `feat/aux-sends-*` branches. It is **not** on the Clouds / Kit Split line, where it was reverted after a file-load freeze traced to its SPI-bus-sharing code — so if your build came from there, there is no AUX submenu to find and no toggle for it in Community Features. Figure 1.1 says which branches carry it.

An aux send, in the mixing-desk sense.

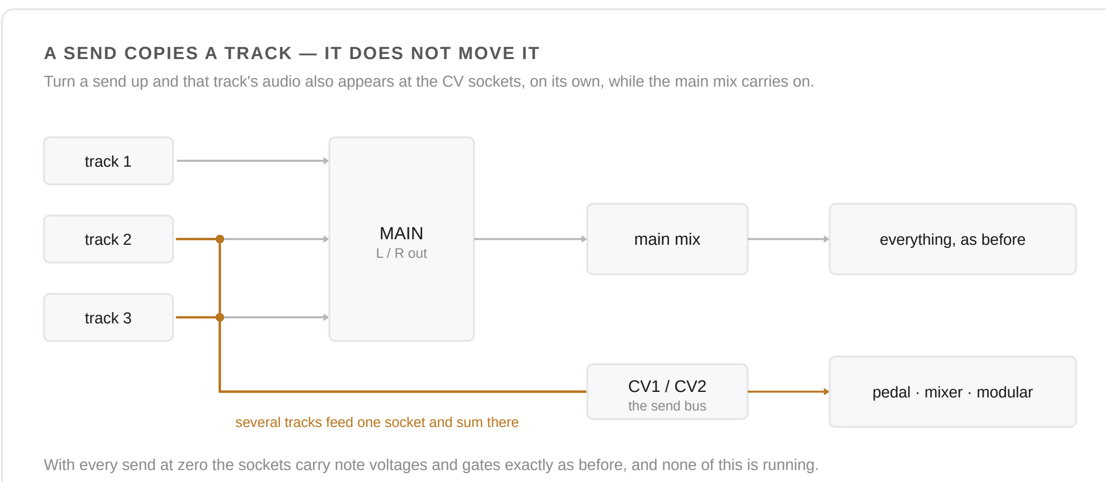


Figure 8.1 — A send copies a track to the sockets without taking it out of the main mix. Several tracks can feed one socket, and they sum there.

Each track gets a send control; turning it up copies that track's audio to the CV outputs at whatever level you set, without taking it out of the main mix. Several tracks can feed the same socket at once, and they sum there.

The sockets keep their ordinary job when nothing is routed to them. With every send at zero they carry note voltages and gates exactly as before, and nothing about the feature is running.

02 Finding it

FINDING THE SENDS

Two homes: the per-track sends, and the global settings for the sockets.

`AUX` on Kit-global, Audio Clip and Sound menus

- CV Send `CV`
- Main `MAIN`

`SETTINGS` › `AUX`

- CV1 Level `CV1L`
- CV2 Level `CV2L`
- Stereo Split `SPLT`

SETTINGS → AUX is deliberately separate from SETTINGS → CV. That one is about pitch voltages; this one is about audio.

Figure 8.2 — Two homes: the per-track sends, and the global settings for the sockets.

Your first minute

#	DO THIS	AND WATCH
1	Open `AUX` on a track and turn **CV Send** up	That track"s audio now also leaves by the CV sockets, without leaving the main mix
2	Count the send controls	One means Stereo Split is on and the pair is a single stereo destination. Two means it is off and they are separate mono buses
3	Turn a second track"s send up too	They sum at the socket, like any aux bus
4	Turn **Main** off on one of them	That track leaves the mains entirely and exists only on the send. Set it from the arranger — elsewhere it edits a copy the playback path does not read
5	Leave the screen alone while it plays	On an OLED unit the display and the CV converter share a bus. A still screen is silent; a scrolling one ticks

AUX the sends and MAIN, per track
on the Kit-global (affect-entire), Audio Clip and Sound menus

SETTINGS → AUX CV1 Level, CV2 Level, Stereo Split — global

The **AUX** submenu appears wherever a track's own parameters live. It is not offered on the Sound menu when that menu is acting as the Drum editor, because a send has to isolate a whole track — a per-drum send has nothing it could do. The kit-wide send, which is the one that works, is on the Kit-global menu instead.

`SETTINGS → AUX` is deliberately separate from `SETTINGS → CV`. That one is about pitch voltages; this one is about audio.

03 The controls

On the track — the `AUX` menu

CONTROL	7SEG	WHAT IT DOES

CV Send
or **CV1 Send**
+ CV2 Send
	`CV`
	`CV1`
	`CV2`
		How much of this track goes to the sockets. One control when the pair is a single stereo destination, two when they are separate mono buses — section 04. Ordinary parameters: automatable, LEARN-able to a knob, reachable by pad and by scroll in Automation View.

Main	`MAIN`	Whether this track still goes to the main outputs. On by default. Turn it off to move a track entirely onto the send — the mixing-desk idea of a pre-fade send with the fader down.

In settings — `SETTINGS → AUX`

CONTROL	7SEG	WHAT IT DOES
CV1 Level **CV2 Level**	`CV1L` `CV2L`	Output level for each socket, 0–50. Each step is 1.2 dB, so the range is even the whole way down; 0 is a true mute rather than the bottom of the taper. The same value the gold knob on the stutter button's lower encoder drives, so the two always agree.
Stereo Split	`SPLT`	Whether the two sockets are one stereo destination or two independent mono ones. Third item in the menu, after the two levels. See 04.

The split is in `SETTINGS` rather than on the track because it describes how your rig is patched, not what one clip is doing. Per-clip it would have allowed one clip to treat the pair as stereo while another treated it as two mono outs, which no actual patch ever means.

MAIN AND WHERE YOU ARE

MAIN takes effect when a clip is playing from the arranger. Reached from an Audio Clip or from a kit in affect-entire, it edits that clip's own copy, which is not the one the playback path reads — so it will not appear to do anything there. If you want a track off the mains, set it from the arranger.

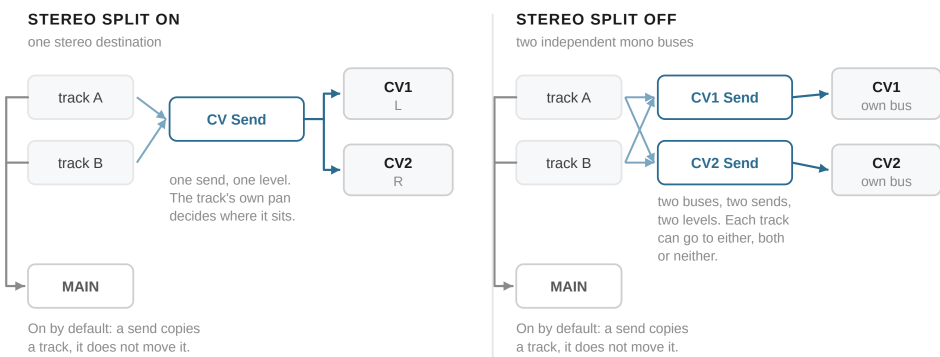
04 One destination, or two

Stereo Split on. The two sockets are the left and right of one stereo destination. There is one send and one level, because there is one place the audio is going, and where a track sits across the pair comes from its own pan — the same pan control it has always had.

Stereo Split off. The sockets are two separate mono aux buses feeding two different boxes. Two buses want two sends and two levels, the same as any mixer, and each track can go to either, both, or neither.

ONE DESTINATION, OR TWO

Stereo Split describes how your rig is patched, which is why it lives in SETTINGS rather than on the track.



Several tracks can feed the same socket at once and they sum there. With every send at zero the sockets carry note voltages and gates exactly as before. The send runs about 16 ms behind the mains — use it to go somewhere, not to come straight back.

Figure 8.3 — Two sockets, read two ways. The number of send controls on a track is how you tell which one you are in: one send means the split is on, two means it is off.

ONLY SEEING ONE SEND CONTROL?

Then Stereo Split is on, and that is correct — one destination gets one send. If you wanted two independent sends, turn it off at `SETTINGS → AUX → Stereo Split` and **CV1 Send** and **CV2 Send** will both appear on the track. The reverse holds too: two sends showing means the split is off.

The same collapse happens to the levels. With the split on, **CV2 Level** has nothing to set and hides itself, because there is one destination and therefore one level.

THE SEND IS ABOUT 16 MS BEHIND THE MAINS

Fine for what a send is normally for — into a pedal, an amp, a mixer channel or a modular case, where it is heard on its own. But if you take the socket's output back into the same mix as the source track, the two copies will comb-filter against each other. Use the send to go somewhere, not to come straight back.

05 On OLED, the screen and the sockets share a bus

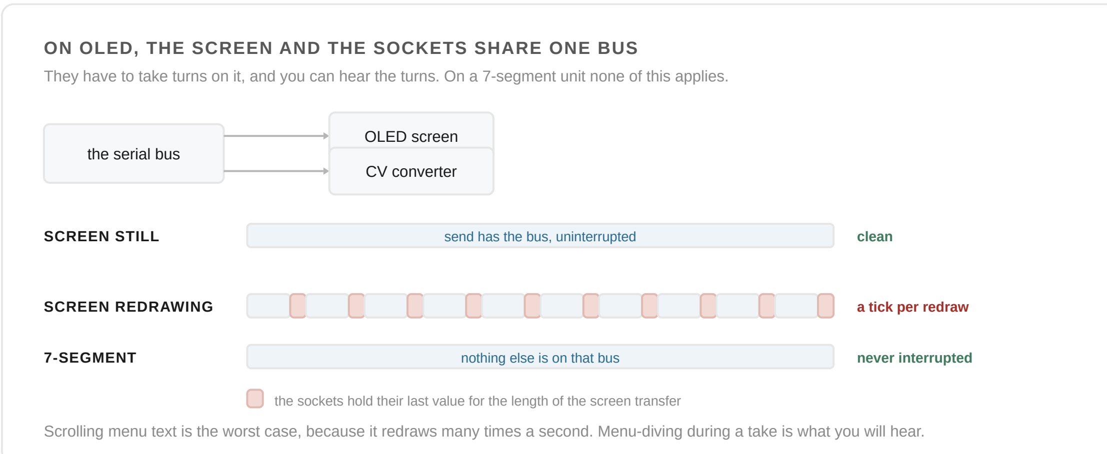


Figure 8.4 — Why a still screen is silent and a scrolling one is not. On a 7-segment unit none of this applies.

Worth knowing before you use this on an OLED machine, because it is audible. The display and the CV converter are wired to the same serial bus there, and they have to take turns on it. On a 7-segment unit none of this applies.

SCREEN STATE	EFFECT ON THE SOCKETS
Still	Nothing at all. The screen is only redrawn when something on it changes, so a screen you are not touching costs nothing and the send is clean.
Redrawing	A tick, once per redraw, as the sockets hold their last value for the length of the screen transfer. Scrolling menu text is the worst case, because it redraws many times a second.
7-segment	Nothing else is on that bus, so the send is never interrupted.

In practice: leave the screen alone while a send is carrying something you care about, and it will be silent. Menu-diving during a take is what you will hear.

A CV/GATE TRACK GOES QUIET WHILE A SEND IS UP

Whenever any send is active, the sockets stop putting out note voltages. This is not a fault — the send bus drives both sockets continuously as audio, so there is no socket left to carry a pitch. Sends and CV/gate are two uses of the same two jacks, and you get one at a time.

06 Reference

SWITCHING IT OFF ENTIRELY

`SETTINGS → COMMUNITY FEATURES → Aux Sends` (`AUX` on 7SEG), on by default. Switching it off hides the AUX submenu and `SETTINGS → AUX` . As with every toggle in this firmware it hides menus rather than stopping processing, so a song already sending audio to the sockets keeps doing so.

PROVENANCE

After sticknobills' Aux Sends, rebuilt here to run on both display types.

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SYNTHSTROM DELUGE · RUDEMENT FIRMWARE

The

Chroma

Operating manual for the Harmonic layout and the Brush

A chord explorer beside an isomorphic visualiser, on one 16-wide surface, plus a brush that stamps a held chord into the piano roll like a note. Pick a chord on the left and it sounds, names itself and lights its shape on the right, where colour and brightness carry the harmony rather than decorating it. This chapter covers the whole surface, the voicing engine behind it, the progression walker, and the four smaller tools that came with it.

Harmonic · Palette · Iso · Brush · BRSH · RCAP

Edition of August 2026

01 What it is

Two seven-wide grids with two control columns wedged between them. On the left, a **palette**: one column per scale degree, stacking upward through an additive richness ladder from a single root note to a 13th. On the right, an **isomorphic play surface** tinted to the key's own colour, where the chord you picked lights as a shape you can play, extend or walk away from.

The two are not a keyboard and a decoration. The palette decides *which* chord; the iso panel shows you where that chord lives across the grid, which of its notes are sounding, and — with the overlays on — every other position it makes available, or how its voices moved from the chord before it.

Everything is scale-relative. There are no chord shapes to memorise and no chord quality is ever written on screen: the Roman numeral carries it, computed from whatever key and scale the song is in.

> TWO HARD REQUIREMENTS
>
> **Scale mode must be on.** The whole layout is built from scale degrees; outside a scale it has nothing to say, and it refuses to open.
>
> **Melodic tracks only.** It will not open on a kit.

02 Getting in

GETTING INTO CHROMA

Harmonic is last in the layout list, so it is one step back from Isomorphic turning anticlockwise.

`hold KEYBOARD` › `turn SELECT` › **`Harmonic`**

└ Scale mode must be on
└ Melodic tracks only — not a kit

On a 7-segment display it is HARM on the 1.3 line and scrolls HARMONIC on 1.2.1.

Figure 9.1 — Two requirements and one gesture.

Hold **KEYBOARD** and turn the **SELECT** encoder to cycle layouts. Harmonic is last in the list, so it is one step back from Isomorphic turning anticlockwise. On OLED it names itself *Harmonic*; on a 7-segment display it is `HARM` on the 1.3 line and scrolls `HARMONIC` on 1.2.1.

Your first minute

#	DO THIS	AND WATCH
1	Press column 0, row 2 — bottom left, third row up	The tonic triad sounds and names itself
2	Look right	The same chord, lit in the column's own colour, with an amber pad an octave below it — that is the bass note
3	Look left	Two or three columns pulsing white on rows 2–5. That is the Calculator saying where this chord usually goes. Press one
4	Hold a chord pad and turn the vertical encoder	The voicing walks while the chord rings. Let go and it springs back
5	Hold LEARN and tap anything	The surface goes silent and every pad tells you what it is

On a scale with fewer than seven notes you get fewer palette columns and the leftovers go black. The Calculator needs exactly seven and stays dark on anything else. Neither is a fault.

03 The surface

Sixteen columns: palette 0–6, palette control 7, iso control 8, iso 9–15. **SWAP** mirrors the whole thing — palette right, iso left — and each control column moves with the grid it drives. Rows are numbered 0 at the bottom, 7 at the top, as the hardware is; every table here reads top-down, the way the column sits under your hand.

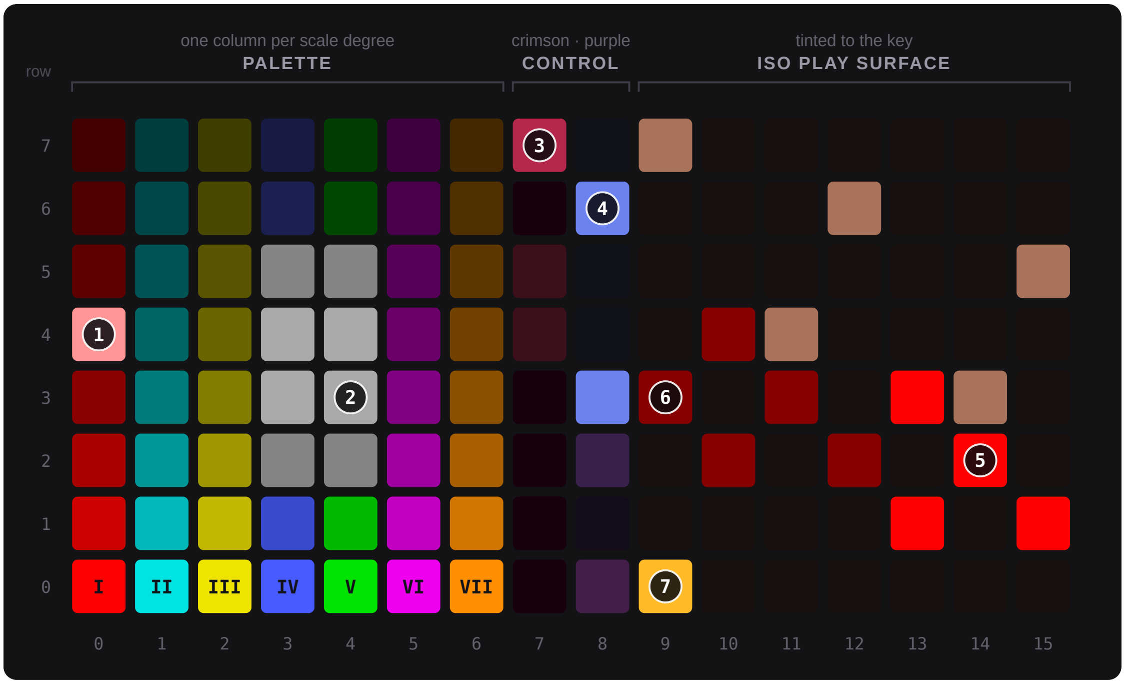


Figure 9.2 — The whole surface, in C major, with the tonic 7th picked. Pad colours here and in every figure below are the values the layout actually renders; brightness is lifted at the dim end so faint pads survive printing, but the relative order is exactly as built.

1. the picked cell — a near-white tint of its own hue
2. the Calculator, pulsing white on rows 2–5, saying where this chord usually goes
3. CALCULATOR, lit — the crimson column shows only what is engaged
4. SHOW CHORD SHAPE, lit in its band's purple
5. a voiced note at its primary (lowest) position — full bright
6. the same note repeating higher up the grid — half bright
7. the bass spotlight, an octave below the voicing — amber

The palette — columns are degrees, rows are richness

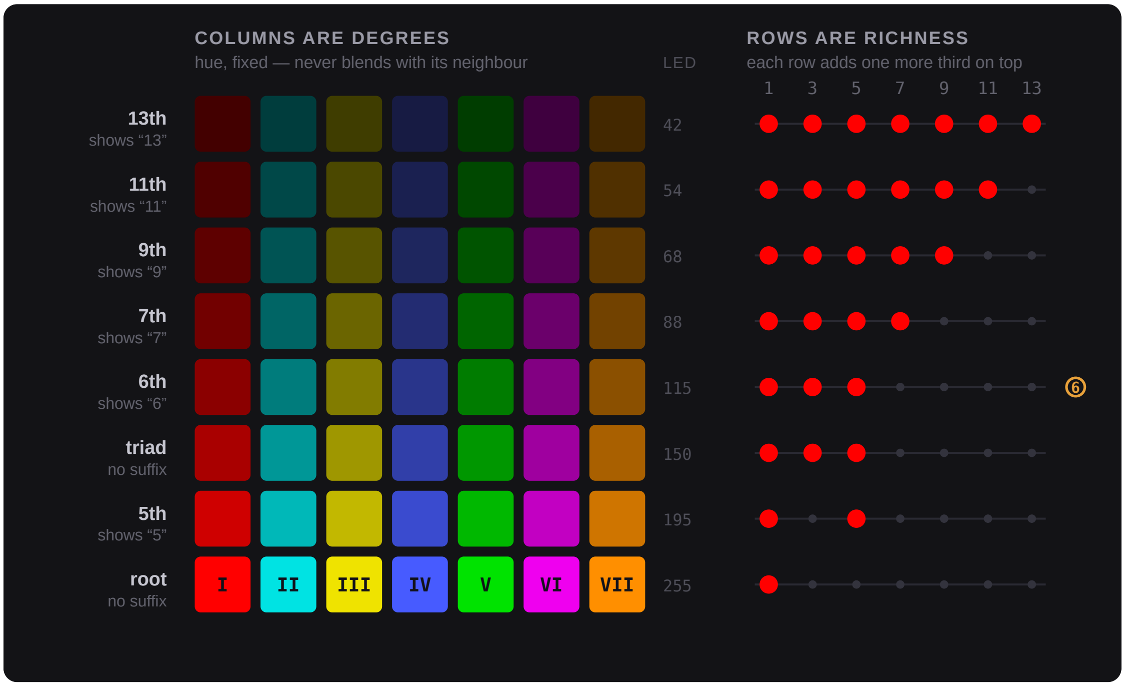


Figure 9.3 — One axis each. Across, seven fixed hues, ordered so no two neighbours blend. Up, the additive ladder — each row stacks one more third, and the LED dims as the chord thickens, so a column reads as getting denser. The 6th row is the odd one: a triad plus a borrowed major 6th, not another third.

ROW	CHORD	NOTES	SUFFIX SHOWN
7	13th	root 3 5 7 9 11 13	13
6	11th	root 3 5 7 9 11	11
5	9th	root 3 5 7 9	9
4	7th	root 3 5 7	7
3	6th	triad + a major 6th, borrowed if the scale only has a ♭6	6
2	**Triad**	root 3 5	—
1	5th	root + 5th	5
0	**Root**	single note — the bass / anchor lane	—

Each column has its own hue — red, cyan, yellow, blue, green, magenta, orange — chosen so neighbours never blend. Brightness fades as you climb, so a column reads as getting denser. A picked cell blows out to a near-white tint of its own hue, so it still pops on the dim upper rows.

The name on screen is absolute then Roman, and the absolute half is only the root plus the suffix above. A I7 in C reads `C7 I7` — not `Cmaj7`, because the layout never writes a quality. A ii triad reads `D ii`.

sus2 and *sus4* are deliberately not on the ladder. They are alterations, made in EDIT mode by toggling the 3rd out and the 2nd or 4th in.

The iso panel — what the colours mean

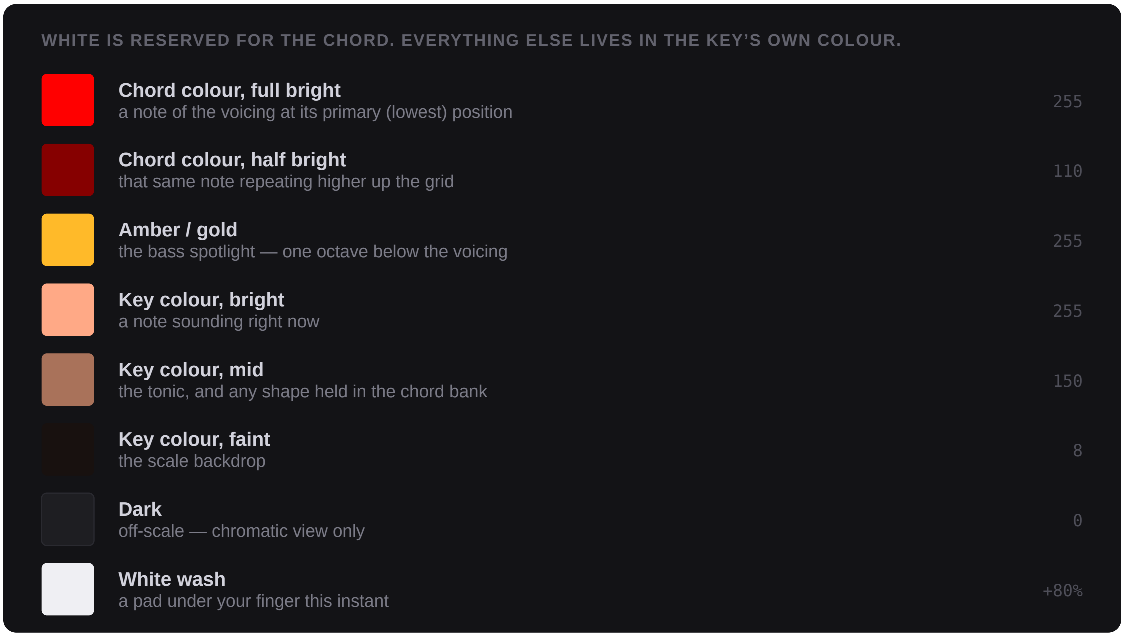


Figure 9.4 — The eight states, drawn rather than described — here in C, whose key colour is a warm coral, with a red (degree I) chord picked. The number on the right is the LED level the firmware asks for, out of 255.

WHAT YOU SEE	WHAT IT MEANS
Chord colour, full bright	A note of the current voicing, at its primary (lowest) position
Chord colour, half bright	That voiced note repeating higher up the grid
Amber / gold	The **bass spotlight** — one octave below the voicing's lowest note. It sits at its true register, so it may be below your current scroll
Key colour, bright	A note sounding right now
Key colour, mid	The tonic, and any shape held in the chord bank
Key colour, faint	The scale backdrop

Dark	Off-scale — chromatic view only
White wash	A pad under your finger this instant

One catch the table cannot show: while a chord is sounding, every occurrence of it goes full bright. The primary/repeat distinction appears when the notes stop, so it teaches you the shape after you play it rather than during.

The **vertical encoder scrolls the iso panel**, octaves 1–8, whenever no chord pad is held. The palette's own octave is separate — the two `OCT` pads on the crimson column. Keeping them apart is the point: voice chords low, play the melody high. Free play on the iso clears the chord selection unless STICKY is on or the octave picker is open.

04 The control columns

The names below are exactly what LEARN says when you tap that pad, so this page can be checked against the instrument. Crimson shapes the chord; purple shapes the surface.

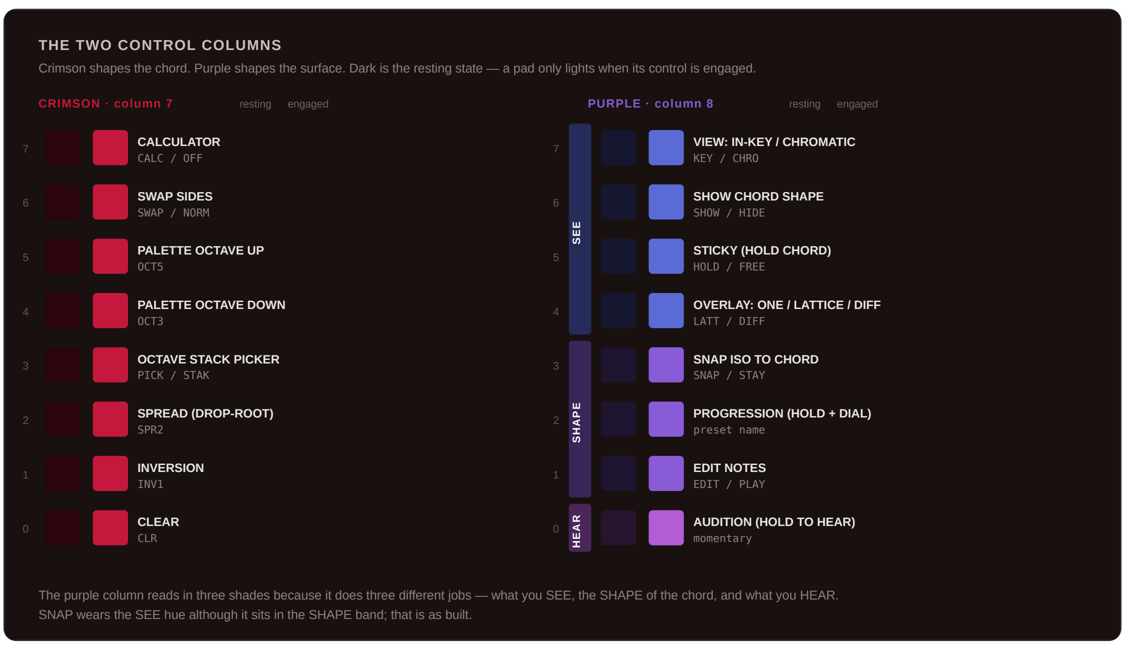


Figure 9.5 — Both control columns, with every row drawn twice — dark as it rests, lit as it engages. The purple column reads in three bands because it does three different jobs.

Palette control — crimson

ROW	CONTROL	WHAT IT DOES	DISPLAY
7	CALCULATOR	Next-chord suggestions on and off	`CALC / OFF`

6	SWAP SIDES	Mirror palette and iso	SWAP / NORM
5	PALETTE OCTAVE UP	Chord register, 1–8	OCT5
4	PALETTE OCTAVE DOWN	The same, downward	OCT3
3	OCTAVE STACK PICKER	Bottom iso row becomes seven octave toggles	PICK / STAK
2	SPREAD (DROP-ROOT)	Cycles 0–3. Silent until you replay the chord	SPR2
1	INVERSION	Root → 1st → 2nd → 3rd. Re-strikes, so you hear it	INV1
0	CLEAR	Clears the selection — or toggles VOICED/BARE while PROG is held	CLR

Iso control — purple

ROW	CONTROL	WHAT IT DOES	DISPLAY
7	VIEW: IN-KEY / CHROMATIC	Switches the iso mapping	KEY / CHRO
6	SHOW CHORD SHAPE	Light the voicing, or a clean grid	SHOW / HIDE
5	STICKY (HOLD CHORD)	Keep the chord through free play	HOLD / FREE
4	OVERLAY: ONE / LATTICE / DIFF	Cycles the three overlays	LATT / DIFF
3	SNAP ISO TO CHORD	On pick, jump the iso to the chord's octave	SNAP / STAY
2	PROGRESSION (HOLD + DIAL)	Momentary — hold it and dial	preset name
1	EDIT NOTES	Iso taps toggle notes in and out of the voicing	EDIT / PLAY
0	AUDITION (HOLD TO HEAR)	Momentary — hold to hear the voicing	—

The three overlays

ONE shows the voicing you picked, one shape.

LATTICE shows every position of every chord tone across the grid, glowing faintly and fading upward, chord tones brighter than extensions — everywhere the chord is available to play.

DIFF shows voice leading. Notes **held** from the previous chord sit steady, notes **arriving** breathe bright, and notes **leaving** ghost in the previous chord's colour, breathing in antiphase. Watch a ii–V–I and you see the voices walk.

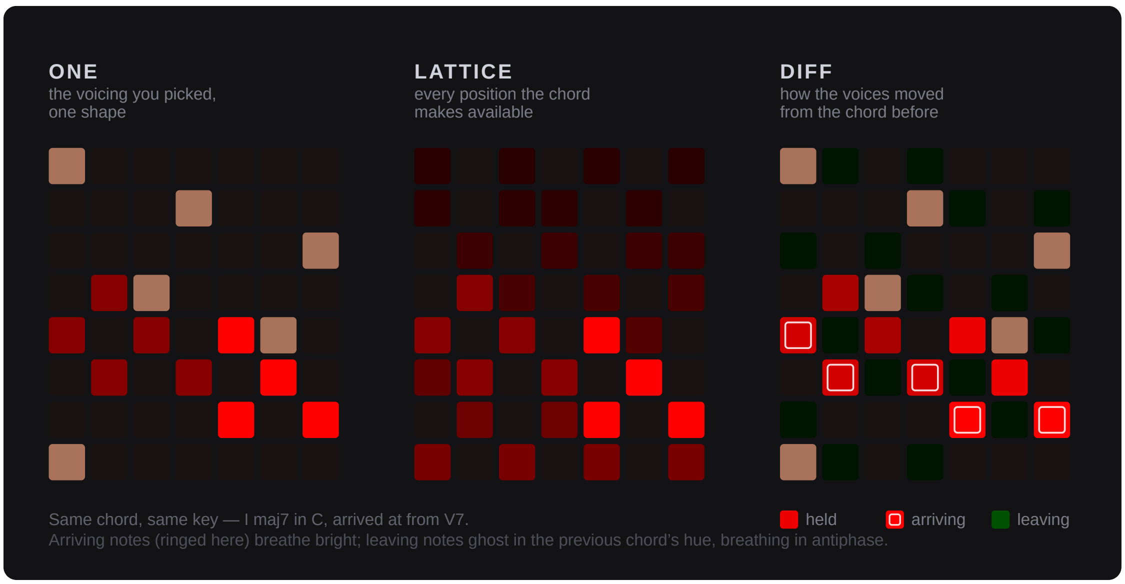


Figure 9.6 — The same chord under each overlay. ONE is the shape you picked; LATTICE glows at every position that chord makes available, chord tones brighter than extensions, fading upward; DIFF drops the lattice and shows the voice leading instead. On the instrument the arriving notes breathe and the leaving ones breathe against them — a still picture can only ring them.

SNAP or STAY, and EDIT NOTES

SNAP, the default, keeps the iso locked to the chord's register so the shape is always visible. **STAY** parks it where you scrolled it — chords low in the left hand, melody high in the right.

EDIT NOTES forces SHOW and STICKY on, stops single taps sounding, and makes each iso pad toggle that note in or out of the voicing. Every edit that lands on a chord the table recognises re-names it; sculpt something it does not know and the display simply keeps the last name. Nothing sounds while you sculpt — tap AUDITION to hear it.

05 The voicing engine

Four transforms, always in this order:

`base chord → INVERSION → SPREAD → WALK → OCTAVE STACK → what you hear`.

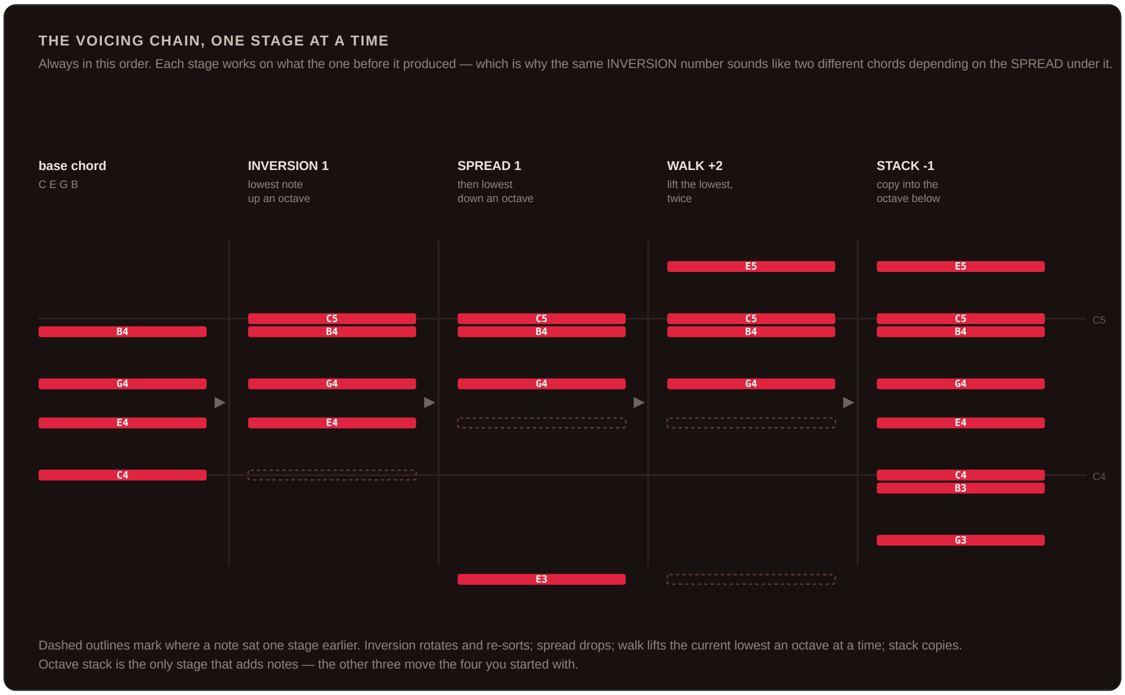


Figure 9.7 — C E G B through the whole chain, one stage at a time, with every note named. Each stage works on what the one before it produced.

STAGE	RANGE	WHAT IT DOES
Inversion	0–3	Rotates: the lowest N notes go up an octave and the chord re-sorts
Spread	0–3	Drops the lowest N notes an octave — drop-root openings
Walk	−24…+24	Walks the whole voicing a note at a time: each step up lifts the current lowest note an octave, each step down drops the current highest
Octave stack	−3…+3	Copies the chord into any of seven octaves, base always on, deduplicated and clamped

The spring-loaded dial

The part worth practising.

GESTURE	RESULT	DISPLAY

Hold a chord pad, turn the vertical encoder	The voicing walks while the chord rings	V3 V-2
Release the chord	Springs back to home	—
Click the vertical encoder	Springs back to home now	VOI0
Hold the encoder in and turn	Dials *where* home is	H4
Hold a chord pad, turn the horizontal encoder	Blooms the octave stack outward: −1, +1, −2, +2, −3, +3	STK2

The octave picker (crimson row 3) turns the bottom row of the iso into seven octave toggles: local columns 0–6 are octaves −3 to +3, column 3 is the base and is always on. Toggling re-strikes the chord so you hear it. Press again and the row goes back to playing notes.

06 Progressions and chord packs

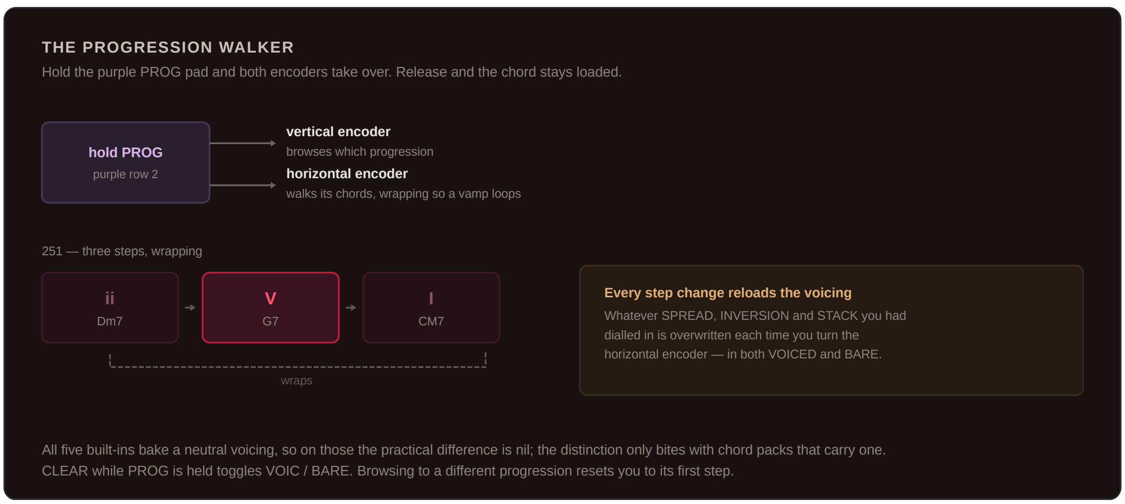


Figure 9.8 — Hold PROG and both encoders change job — one browses the progressions, the other walks the chords.

Hold the purple **PROG** pad and the encoders take over: **vertical** browses which progression, **horizontal** walks its chords, wrapping so a vamp loops. Holding sustains the current step; release and the chord stays loaded. CLEAR while PROG is held toggles `VOIC / BARE`.

> **EVERY STEP CHANGE RELOADS THE VOICING**
>
> Whatever SPREAD, INVERSION and STACK you had dialled in is overwritten each time you turn the horizontal encoder — in both VOICED and BARE. All five built-ins bake a neutral voicing, so on those the practical difference is nil; the distinction only bites with chord packs that carry one.

The five built-ins, all loaded as 7th chords rather than triads

NAME	DEGREES
`251`	ii – V – I
`1564`	I – V – vi – IV
`EPIC`	i – VI – III – VII
`ANDL`	Andalusian: i – VII – VI – V
`DORI`	Dorian vamp: i – IV

They are scale-relative, so they resolve into whatever key and scale the song is in. Browsing to a different progression resets you to its first step.

Chord packs

Put `.chordpack` files in a `CHORDS/` folder at the root of the card and their progressions appear after the built-ins. They are scanned lazily on the first PROG press, never at boot, so a malformed pack cannot brick the instrument — it simply does not load.

FIELD	VALUES
`degree`	1-based, clamped 1–7
`rich`	`root` or 1, 5, triad, 6, 7, 9, 11, 13. Anything else becomes 7
`octup / octdn`	0–3 octave layers above / below the base
`spread / inv`	0–3 each

Limits: 16 pack files, 48 progressions including the built-ins, 8 steps each, and 7-character names. Unknown tags are skipped, so your own metadata is safe. A progression with no name is auto-named `PACK`; one that yields no valid steps is dropped silently.

07 The Harmonic Brush

Capture a chord anywhere, stamp it into the piano roll like a note. The reason it works from everywhere is that it captures **the notes currently sounding**, not a chord index — the Harmonic palette, Chord, Chord Library, or four notes you are simply holding down on the Piano layout are all the same to it.

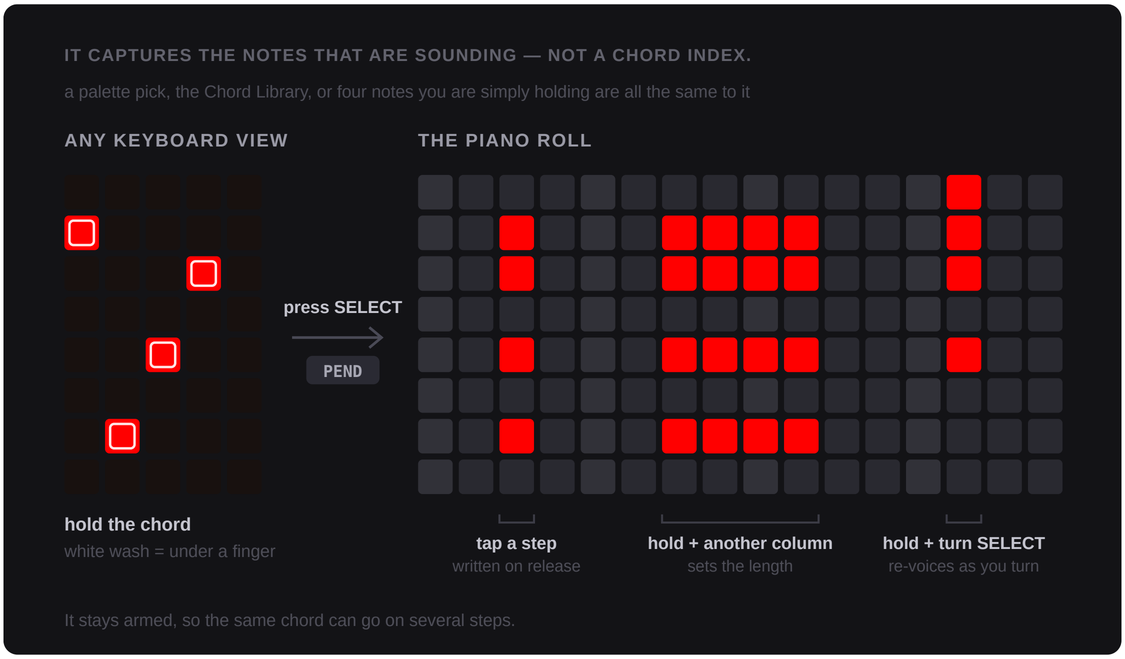


Figure 9.9 — Arm it on whatever is sounding, then stamp. Length comes from holding the placement pad and pressing another column; re-voicing comes from holding it and turning SELECT.

	STEP	DISPLAY
Arm	In a keyboard view, hold the chord and press the SELECT encoder. The brush survives the switch to the piano roll	PEND
Place	Press KEYBOARD to reach the piano roll, then tap a step. The chord auditions the moment you press and is written on release	PLCD
Length	Hold the start step and press any other column — either direction, last press wins	—
Re-voice	While holding the placement pad, turn SELECT. Clockwise lifts the lowest note an octave, anticlockwise drops the highest, re-auditioning as you turn	VUP / VDN
Again	It stays armed, so the same chord can go on several steps	—
Clear	Press SELECT while armed but not holding notes	CLR

It disarms when you switch layout, when you click SELECT, or when you leave the piano roll for any other screen — **which includes going back to Keyboard View**. Arm it there, come here. Each stamped chord is one undo step. Melodic clips only.

08 The three smaller tools

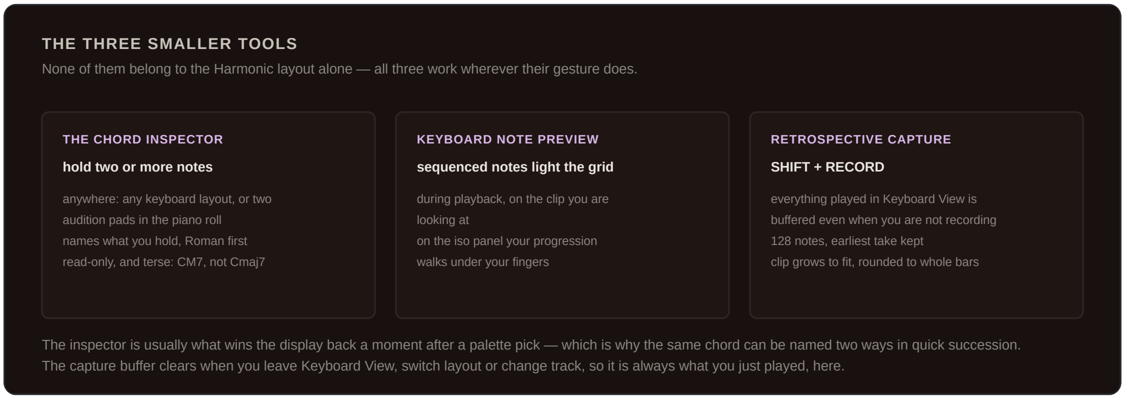


Figure 9.10 — All three work wherever their gesture does, not only on the Harmonic layout.

The chord inspector

Hold two or more notes anywhere — any keyboard layout, or two or more audition pads in the piano roll — and the display names the chord instead of the note. It matches what you hold against the chord table and spells it to the song's key, Roman first: `IM7 CM7`. The table's names are terse, so it reads `CM7`, not `Cmaj7`. Fewer than two notes, or a set the table does not recognise, and you get the ordinary single-note readout. It is read-only, and it is also what usually wins the display a moment after a palette pick — which is why the same chord can be named two ways in quick succession.

Keyboard note preview

With this on — it is by default — sequenced notes light the keyboard grid *during playback*. On the Harmonic iso panel that means your clip plays back as moving shapes on the visualiser: the progression walks under your fingers. It lights only the clip you are looking at, and clears on every path out of a note, so loops and re-triggers cannot leave a pad stuck lit.

Retrospective capture

Every note you play in Keyboard View is buffered in the background, **even when you are not recording**. SHIFT + RECORD dumps the buffer into the current clip. It holds 128 notes and keeps the earliest take rather than overwriting when it fills. Timing is kept in audio samples and converted at dump time, so it works with the transport stopped. The clip grows to fit, rounded to whole bars — two bars noodled over a one-bar loop becomes a real two-bar clip instead of folding back on itself. The display shows how many notes landed; `NONE` means the buffer was empty and `MELODIC ONLY` means you are on a kit. The buffer clears when you leave Keyboard View, switch layout or change track, so it is always *what I just played, here*.

09 Reference

Display codes

Figure 9.11 — Every code the layout can put on a 7-segment display.

CODE	MEANING	CODE	MEANING
`CALC / OFF`	Calculator	`KEY / CHRO`	Iso view
`SWAP / NORM`	Handedness	`SHOW / HIDE`	Chord shape
`OCT3`	Palette octave	`HOLD / FREE`	Sticky
`PICK / STAK`	Octave picker	`LATT DIFF ONE`	Overlay
`SPR2`	Spread	`SNAP / STAY`	Iso follows chord

`INV1`	Inversion	`EDIT / PLAY`	Edit mode
`STK2`	Stack layers	`VOIC / BARE`	Progression voicing
`V3 V-2`	Voicing walk	`PEND`	Brush armed
`VOI0`	Sprung back to home	`PLCD`	Chord placed
`H4`	Voicing home	`VUP / VDN`	Brush re-voice
`CLR`	Cleared	`NONE`	Capture buffer empty
`BANK / EMPTY`	LEARN on a bank pad	`MELODIC ONLY`	Capture on a kit

Defaults on first entry

Palette octave 3, iso octave 3, in-key view, Calculator on, chord shown, SNAP on, sticky off, both overlays off, no swap, edit mode off, octave picker closed, stack at the base octave only, spread 0, inversion 0, walk 0, home 0, progressions VOICED.

Sidebar

Stays at the factory velocity / mod default. The chord bank is opt-in: hold a sidebar column and turn the vertical encoder to switch it to chord memory. The `CHORD` column function is blocked in this layout.

> AVAILABLE ON BOTH FIRMWARE LINES
>
> Chroma ships as three commits on each line — `feat/chroma-13` and `feat/chroma-12` — and merged into each line's build as `integration/chroma-13` and `integration/chroma-12`. The layout source is byte-identical between them.
>
> Three differences are visible from the front panel: the 7-segment layout name (`HARM` on 1.3, scrolling `HARMONIC` on 1.2.1), the Piano layout, which does not exist on the 1.2.1 line at all, and the number of new strings the port needed. Everything else behaves the same.

Toggles

Three of the nine Community Features toggles belong to this chapter — **Chord Brush** (`BRSH`), **Keyboard Note Preview** (`PREV`) and **Retrospective Capture** (`RCAP`), all on by default. Chapter 01 lists what each one hides. The Harmonic layout itself has no toggle: it is always there.

PROVENANCE

Original to this fork, 2026. The layout, the voicing engine, the progression walker and the Harmonic Brush are all new here; the chord table it names against is the Deluge's own. Source: `src/deluge/gui/ui/keyboard/layout/harmonic.cpp`, `src/deluge/gui/ui/keyboard/chord_library.cpp`.

↑ Back to contents

The

MIDI Implementation Chart

Reference for every CC assignment across this fork

Every default MIDI CC baked into this firmware, and every added control that has none, in one place. The nine earlier chapters describe each feature's controls; this one answers a single cross-cutting question for each of them — what happens when a CC arrives, or when you reach for LEARN. Figures are read from source, not measured on a controller, and are called out by firmware line where the two differ.

CC · FOLLOW · LEARN

Edition of August 2026

01 How to read this chart

Three separate things determine whether a MIDI CC does anything to a control, and this chart tracks all three:

Default CC (MIDI Follow). A fixed CC baked into `defaultParamToCCMapping`, the table MIDI Follow mode loads from the moment `SETTINGS/MIDIFollow.XML` is first created. It works the instant you set a Follow channel — no LEARN required — and only for the context that control belongs to (per-voice, per-track, or song-global). Hand-editing the XML overrides it per parameter.

Shortcut grid. A control needs a fixed pad position in one of the three grid-sized tables in `param.h` (patched, unpatched-non-global, unpatched-global) before it can have a default

CC at all — MIDI Follow resolves an incoming CC back to a parameter by scanning the CC table for that same grid position. The same position also puts a control in the Automation View parameter list. No grid position means no default CC, full stop.

MIDI Learn. The older, separate mechanism: hold LEARN and move a knob on your controller, and that CC drives this one control in this one song, regardless of grid position or MIDI Follow. Any real modulation-capable parameter supports it. A menu selection, a momentary action, or a bare on/off latch that never became a parameter (Wisps' Mode, Kit Split's split action) does not.

> **WHERE THIS IS DOCUMENTED UPSTREAM**
>
> MIDI Follow mode itself — channels, feedback, the XML format — is covered in full in `docs/features/midi_follow_mode.md`, unchanged by this fork. This chapter only adds what that page can't: which of the CCs it lists belong to Rudement's own additions, and which of this fork's controls it silently leaves out.

02 The Gristleator — chapter 05

All ten controls, including the On switch, sit on the shortcut grid and carry a default CC. Rate through Level use **102–109**, chosen because that range is undefined in the MIDI spec and unused elsewhere in the table; Dirt takes 110, next door. On took 89 specifically to match 1.3 — see the note under chapter 03 below.

CONTROL	7SEG	DEFAULT CC
On	`GON`	89
Rate	`GRAT`	102
Depth	`GDEP`	103
Shape	`GSHP`	104
Bias	`GBIA`	105
Mode	`GMOD`	106
Freq	`GFRQ`	107
Reso	`GRES`	108
Level	`GLVL`	109

Dirt	`GDRT`	110

A CC IS INSEPARABLE FROM A GRID POSITION HERE

Straight from the source comment in `midi_follow.cpp` : this table is indexed by grid position, and MidiFollow resolves a received CC back to a parameter by scanning it for a position — not the other way round. Removing the Gristleator's pads from the shortcut tables would silently remove these ten CCs too, not just the on-screen shortcuts.

03 The Four-Band EQ — chapter 02

All eight controls — the two stock shelves included — sit on the shortcut grid. The two mid bells share their row with the Gristleator's On switch, which is what forced the one change worth flagging: **Low mid freq moved from CC 89 to CC 90** so that On could take 89 and match 1.3. Nothing else on this row shifted.

CONTROL	7SEG	DEFAULT CC
Bass amount	`BASS`	86
Bass freq	`Bfrq`	84
Low mid amount	`LOW MID`	88
Low mid freq	`LMfq`	90 (was 89)
High mid amount	`HIGH MID`	111
High mid freq	`HMfq`	112
Treble amount	`TREBLE`	87
Treble freq	`Tfrq`	85

Bass and treble are stock Deluge and their CCs predate this fork; included here for completeness, since a CC collision doesn't care which chapter documented a control first.

04 Sear — chapter 06

Sear (the drive) is a patched parameter and sits on the shortcut grid: default CC **22**. **STon** (the tone control) is unpatched and off the grid entirely — no default CC, not in the Automation View shortcut list. It is still an ordinary modulation-capable parameter, so

MIDI Learn (hold LEARN, move a knob) works on it exactly as it would on any other unpatched control; there is simply nothing waiting for it out of the box.

CONTROL	7SEG	DEFAULT CC
Sear	`SEAR`	22
STon	`STon`	—

Sear's neighbour on the same shortcut row is Fold, a wavefolder that predates this fork and isn't one of its chapters — it holds CC 19, immediately below Sear's 22, for anyone auditing that region of the table.

05 Bouls — chapter 03

Harmonics, Timbre and Morph were never given CCs of their own — as chapter 03 explains, they reuse three parameter slots that already existed for other oscillator types, and inherit whatever CC already controlled that slot. There is nothing Bouls-specific to remap: point a controller at Oscillator 1's Pulse Width and Wave Index, or Oscillator 2's Pulse Width, and Bouls answers.

BOULS CONTROL	BORROWED SLOT	DEFAULT CC
Harmonics	Osc 1 Pulse Width	23
Timbre	Osc 1 Wave Index	25
Morph	Osc 2 Pulse Width	28

CC 25 is the exception worth knowing about. The wave-index parameter Timbre borrows is centred, and the engine treats everything below its midpoint as zero. The Bouls menus compensate for that; a CC does not. Timbre therefore only starts moving at CC 25 value 64, and the whole of a controller's lower travel does nothing. CC 23 and CC 28 are one-sided parameters and behave normally across their full range.

Model, Low-pass gate, LPG decay/colour and Aux output are menu selections, not modulatable parameters — no CC of any kind reaches them.

06 Wisps — chapter 04

None of Wisps' nine continuous parameters sit on a shortcut grid, so none has a default CC and none appears in the Automation View parameter list or on the shortcut grid itself

— chapter 04 says as much. Every one of them is still an ordinary automatable parameter underneath, so MIDI Learn works control by control, and a fixed CC can be hand-written into `SETTINGS/MIDIFollow.XML` for any of them; the tag is there with a value of 255 (unmapped) waiting to be edited.

CONTROL	7SEG	DEFAULT CC	MIDI LEARN
Blend	`Blnd`	—	Yes
Position	`Posn`	—	Yes
Size	`Size`	—	Yes
Pitch	`Ptch`	—	Yes
Density	`Dens`	—	Yes
Texture	`Text`	—	Yes
Spread	`Sprd`	—	Yes
Feedback	`Fdbk`	—	Yes
Reverb	`Verb`	—	Yes
Mode	`DUST`	—	No — not a parameter
Freeze	—	—	No — not a parameter

07 Aux Sends — chapter 08

Chapter 08 states that CV Send is “automatable, LEARN-able to a knob, reachable by pad and by scroll in Automation View” — meaning it does hold a shortcut-grid position, on both firmware lines, and is therefore MIDI Follow-eligible in principle. This edition has not independently confirmed a default CC for it on either line, and does not print one here rather than guess.

> **CHECK YOUR BRANCH BEFORE YOU CHECK THE CC**
>
> Aux Sends is not on every line — it lives on the `feat/aux-sends-*` branches and was reverted off the Clouds / Kit Split line after a file-load freeze traced to its SPI-bus-sharing code. Chapter 01, figure 1.1, says which branches carry it. On a build without it there is nothing here to assign.

08 Kit Split and Chroma — chapters 07 & 09

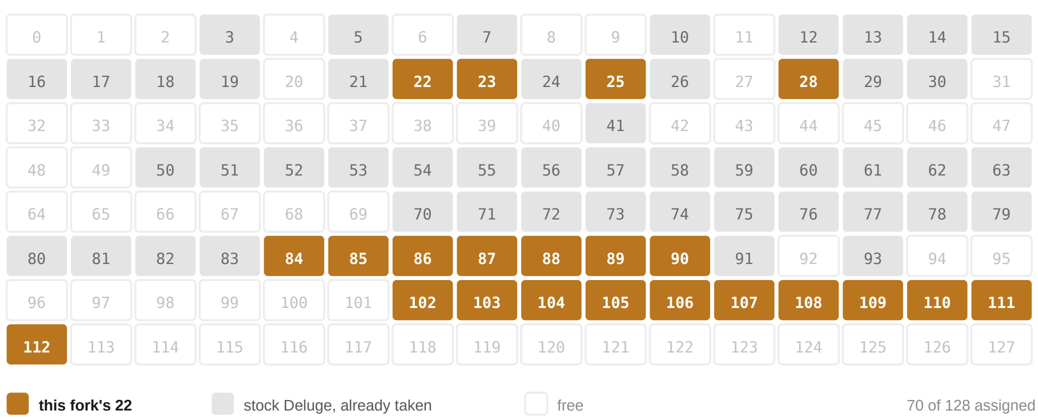
Kit Split is a one-shot menu action, not a parameter of any kind — it has no CC, no shortcut-grid position, and nothing to MIDI-learn. Nothing about it belongs in a CC chart, and it is listed here only to say so explicitly.

Chroma is a performance surface, not a bank of automatable parameters: the palette and iso grids pick and voice chords locally, and none of that is exposed as a modulation destination a MIDI CC could drive. The one place chapter 09 uses the word LEARN is the chord-bank sidebar (hold a sidebar column, turn the vertical encoder) — that binds a physical pad column to chord memory, and has nothing to do with incoming MIDI CCs. Neither feature appears in the reference table below.

09 Reference — every default CC, by number

EVERY DEFAULT CC, AND EVERY ONE STILL FREE

The whole of defaultParamToCCMapping laid out 0-127. Check here before you point a controller at anything.



The Gristleator sits at 102-110 because that range is undefined in the MIDI spec and unused elsewhere in the table.
Low mid freq moved from 89 to 90 so the Gristleator's On could take 89 and match the 1.3 line.
A CC here is inseparable from a shortcut-grid position: remove the pads and the CCs go with them.

Figure 10.1 — The whole CC space at once. Orange is this fork, grey is stock Deluge, outline is free — the fast way to check for a collision before you assign anything new.

Every fixed CC this fork's additions occupy, in one sorted list — the fast way to check for a collision before assigning something new to a controller.

CC	CONTROL	CHAPTER
22	Sear	06 · Sear

23	Bouls · Harmonics	03 · Bouls
25	Bouls · Timbre	03 · Bouls
28	Bouls · Morph	03 · Bouls
84	EQ · Bass freq	02 · Four-Band EQ
85	EQ · Treble freq	02 · Four-Band EQ
86	EQ · Bass amount	02 · Four-Band EQ
87	EQ · Treble amount	02 · Four-Band EQ
88	EQ · Low mid amount	02 · Four-Band EQ
89	Gristleator · On	05 · The Gristleator
90	EQ · Low mid freq	02 · Four-Band EQ
102	Gristleator · Rate	05 · The Gristleator
103	Gristleator · Depth	05 · The Gristleator
104	Gristleator · Shape	05 · The Gristleator
105	Gristleator · Bias	05 · The Gristleator
106	Gristleator · Mode	05 · The Gristleator
107	Gristleator · Freq	05 · The Gristleator
108	Gristleator · Reso	05 · The Gristleator
109	Gristleator · Level	05 · The Gristleator
110	Gristleator · Dirt	05 · The Gristleator
111	EQ · High mid amount	02 · Four-Band EQ
112	EQ · High mid freq	02 · Four-Band EQ

> **PROVENANCE**
>
> Every CC above is read directly from `src/deluge/io/midi/midi_follow.cpp` (the `defaultParamToCCMapping` table) and `src/deluge/modulation/params/param.h` (the three shortcut-grid tables), cross-referenced by grid position, against `feat/clouds-clean-entry` — the tip of the 1.2.1 line as this edition went out. STon, all nine Wisps parameters, and Aux Sends are omitted deliberately — see their chapters above for why.

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